PRACTICAL JOURNAL

in

Advanced Artificial Intelligence
&
Machine Learning

Submitted to

Laxman Devram Sonawane College, Kalyan (W) 421301

in partial fulfillment for the award of the degree of

Master of Science in Information Technology

(Affiliated to Mumbai University)

Submitted by

TEJAS ANIL KAPADE

Under the guidance of

Mrs. Sabina Ansari (Advanced Artificial Intelligence)

&

Dr. Priyanka Pawar (Machine Learning)

Department of Information Technology

Kalyan, Maharashtra

Academic Year 2025-26

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The Kalyan Wholesale Merchants Education Society's

Laxman Devram Sonawane College, Kalyan (W) 421301

Department of Information Technology
Masters of Science – Part II

Certificate

This is to certify that **Mr. Tejas Anil Kapade**, Seat number______________, studying in Masters of Science in Information Technology Part II , Semester II has satisfactorily completed the practical of "Advanced Artificial Intelligence " as prescribed by University of Mumbai, during the academic year 2025-2026

Subject In-charge	Coordinator In-charge	External Examiner

College Seal

ADVANCED ARTIFICIAL INTELLIGENCE

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Practical 1

Aim : Implementing advanced deep learning algorithms such as convolutional neural networks (CNNs) or recurrent neural networks (RNNs) using Python libraries like TensorFlow or PyTorch

Code :

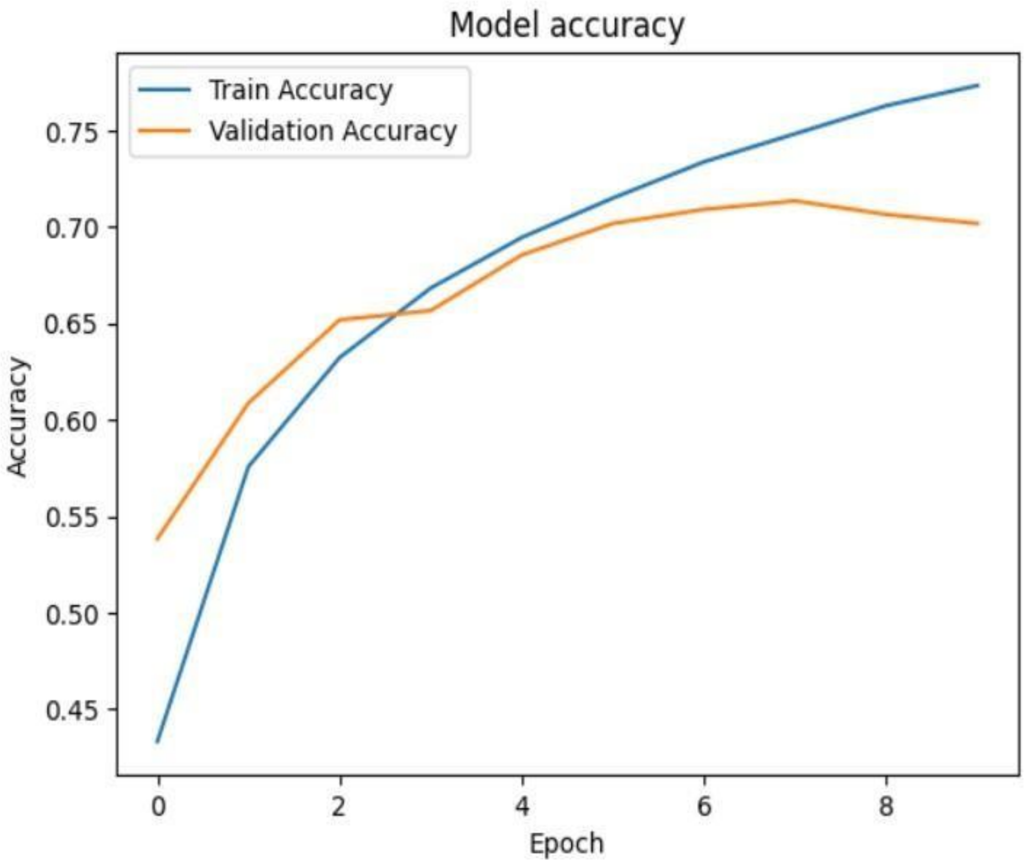
```
import tensorflow as tf
from tensorflow.keras import layers, models import
matplotlib.pyplot as plt

# Load and preprocess the CIFAR-10 dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.cifar10.load_data()
x_train, x_test = x_train / 255.0, x_test / 255.0 # Normalize pixel values to [0, 1]

# Build the CNN model model = models.Sequential([
layers.Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)),
layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'), layers.MaxPooling2D((2,
    2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.Flatten(),              layers.Dense(64,
    activation='relu'),
    layers.Dense(10, activation='softmax') # 10 classes for CIFAR-10
])

# Compile the model
```

```
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train the model
history = model.fit(x_train, y_train, epochs=10, validation_data=(x_test, y_test))

# Evaluate the model test_loss, test_acc =
model.evaluate(x_test, y_test, verbose=2) print(f'\nTest
accuracy: {test_acc}')

# Plot training & validation accuracy values
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation
Accuracy') plt.title('Model accuracy') plt.ylabel('Accuracy')
plt.xlabel('Epoch') plt.legend() plt.show()

import torch import
torch.nn as nn import
torch.optim as optim import
torchvision
import torchvision.transforms as transforms import
matplotlib.pyplot as plt

# Define transformations for the training and testing data
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5)),
])

# Load the CIFAR-10 dataset
```

trainset = torchvision

OUTPUT :

```
Test accuracy: 0.7017999887466431
```



Practical 2

Aim : Building a natural language processing (NLP) model for sentiment analysis or text classification.

Code :

```
!pip install tensorflow matplotlib


import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.preprocessing.text import Tokenizer from
tensorflow.keras.preprocessing.sequence import pad_sequences from
tensorflow.keras.datasets import imdb
import matplotlib.pyplot as plt

# Step 1: Load the IMDB dataset num_words = 10000 # Use the
top 10,000 words in the vocabulary
(x_train, y_train), (x_test, y_test) = imdb.load_data(num_words=num_words)

# Step 2: Explore the dataset
print(f"Number of training samples: {len(x_train)}")
print(f"Number of test samples: {len(x_test)}")
print(f"Sample review (tokenized): {x_train[0]}")
print(f"Label (0 = negative, 1 = positive): {y_train[0]}")

# Step 3: Decode a sample review word_index
= imdb.get_word_index()
reverse_word_index = {value: key for key, value in word_index.items()} decoded_review
= " ".join([reverse_word_index.get(i - 3, "?") for i in x_train[0]]) print(f"Decoded review:
{decoded_review}")
```

 # Step 4: Pad sequences maxlen = 200 # Limit each review to 200 words x_train = pad_sequences(x_train, maxlen=maxlen) x_test = pad_sequences(x_test, maxlen=maxlen)

 # Step 5: Define the model model
= models.Sequential([
layers.Embedding(input_dim=num_words, output_dim=32, input_length=maxlen), layers.LSTM(32), # Use an LSTM layer for capturing sequential dependencies layers.Dense(1, activation='sigmoid') # Output layer for binary classification
])

 # Step 6: Compile the model model.compile(optimizer='adam',
loss='binary_crossentropy',
metrics=['accuracy'])

 # Step 7: Display the model architecture
model.summary()

 # Step 8: Train the model
history = model.fit(x_train, y_train, epochs=5, batch_size=64, validation_split=0.2)

 # Step 9: Evaluate the model
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=2) print(f"Test Accuracy: {test_acc}")

 # Step 10: Plot training history plt.figure(figsize=(12,
4))

 # Accuracy plot plt.subplot(1,

2, 1)

plt.plot(history.history['accuracy'], label='Train Accuracy')

plt.plot(history.history['val_accuracy'], label='Validation

Accuracy') plt.xlabel('Epoch') plt.ylabel('Accuracy') plt.legend()

plt.title('Model Accuracy')

 # Loss plot plt.subplot(1,

2, 2)

plt.plot(history.history['loss'], label='Train Loss')

plt.plot(history.history['val_loss'], label='Validation

Loss') plt.xlabel('Epoch') plt.ylabel('Loss') plt.legend()

plt.title('Model Loss') plt.show()

OUTPUT :

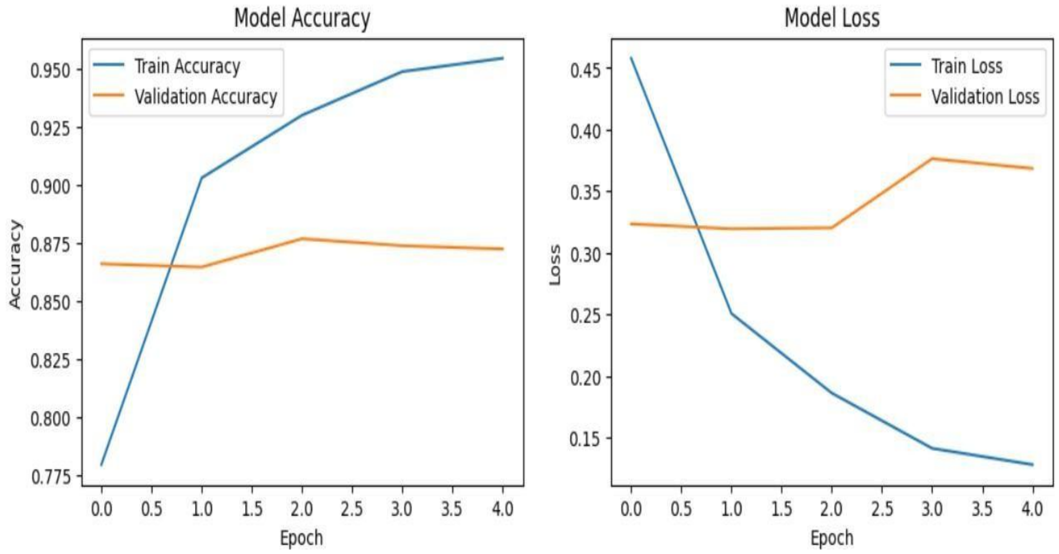
```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz
17464789/17464789 ━━━━━━━━━━━━━━━━ 0s 0us/step
Number of training samples: 25000
Number of test samples: 25000
Sample review (tokenized): [1, 14, 22, 16, 43, 530, 973, 1622, 1385, 65, 458, 4468, 66, 3941, 4, 173, 36, 256, 5, 25, 100, 43, 838, 112, 50, 670, 2, 9, 35, 480, 284, 5, 150, 4, 172,
Label (0 = negative, 1 = positive): 1
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	?	0 (unbuilt)
lstm (LSTM)	?	0 (unbuilt)
dense (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)

Test Accuracy: 0.8611999750137329



Practical 3

Aim : Creating a chatbot using advanced techniques like transformer models.

Code :

```
!pip install transformers torch

from transformers import AutoModelForCausalLM, AutoTokenizer import
torch
```

```
# Step 1: Load Pre-trained Model and Tokenizer
print("Loading the DialoGPT model...") model_name =
"microsoft/DialoGPT-medium" tokenizer =
AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(model_name)

# Step 2: Initialize Chat History
chat_history_ids = None step = 0

# Step 3: Chat with the User
print("Chatbot is ready! Type 'exit' to end the chat.\n")

while True: # User input
   user_input = input("You: ") if
   user_input.lower() == 'exit':
   print("Chatbot: Goodbye!")
   break
```

 # Encode the user input and add it to the chat history new_input_ids = tokenizer.encode(user_input + tokenizer.eos_token, return_tensors='pt') chat_history_ids = torch.cat([chat_history_ids, new_input_ids], dim=-1) if step > 0 else new_input_ids

 # Generate a response using the model response_ids = model.generate(chat_history_ids, max_length=1000, pad_token_id=tokenizer.eos_token_id) response = tokenizer.decode(response_ids[:, chat_history_ids.shape[-1]:][0], skip_special_tokens=True)

 # Display the response

print(f"Chatbot: {response}")

step += 1

Output :

```
You: Hello
The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may observe unexpected behavior. Please pass your input
Chatbot: Hello! :D
You: How are you ?
Chatbot: I'm good, how are you?
You: What is your Name ?
Chatbot: I'm not sure.
You: What is your Qualification ?
Chatbot: I'm a human being.
You: What is your running speed ?
Chatbot: What is your favorite color?
You: Red & What is yours ?
Chatbot: I'm not sure.
You: exit
Chatbot: Goodbye!
```

<u>Practical 4</u>

<u>Aim :</u> Developing a recommendation system using collaborative filtering or deep learning approaches.

Code :

Step 1: Install Required Libraries

Run the following command to install the necessary libraries: pip

install tensorflow numpy pandas matplotlib

Step 2: Download the Dataset

Download the MovieLens 100K dataset from grouplens.org/datasets/movielens. Extract the dataset into a folder.

Alternatively, the code below assumes that the u.data file is in the ml-100k folder.

Step 3: Python Code for the Recommendation System

import pandas as pd import numpy as np import

tensorflow as tf

from sklearn.model_selection import train_test_split import

matplotlib.pyplot as plt

 # Step 1: Load and preprocess the dataset file_path

= "ml-100k/u.data"

column_names = ['user_id', 'item_id', 'rating', 'timestamp'] data

= pd.read_csv(file_path, sep='\t', names=column_names)

Normalize user IDs and item IDs to start from 0 data['user_id']
-= 1

data['item_id'] -= 1

 # Extract the number of users and items

num_users = data['user_id'].max() + 1

num_items = data['item_id'].max() + 1

print(f"Number of users: {num_users}, Number of items: {num_items}")

 # Step 2: Split data into training and testing

train_data, test_data = train_test_split(data, test_size=0.2, random_state=42)

 # Step 3: Create TensorFlow datasets def

create_tf_dataset(df):

    users = tf.constant(df['user_id'].values, dtype=tf.int32)

    items = tf.constant(df['item_id'].values, dtype=tf.int32)

    ratings = tf.constant(df['rating'].values, dtype=tf.float32)

   return tf.data.Dataset.from_tensor_slices(((users, items), ratings)).shuffle(1024).batch(32)

train_dataset = create_tf_dataset(train_data) test_dataset

= create_tf_dataset(test_data)

 # Step 4: Define the Recommendation Model class

MatrixFactorizationModel(tf.keras.Model):

    def _init_(self, num_users, num_items, embedding_dim=50):

        super()._init_() self.user_embedding = tf.keras.layers.Embedding(num_users,

        embedding_dim) self.item_embedding =

        tf.keras.layers.Embedding(num_items, embedding_dim)

    def call(self, inputs):

user_vector = self.user_embedding(inputs[0]) item_vector

= self.item_embedding(inputs[1])

dot_product = tf.reduce_sum(user_vector * item_vector, axis=1)
return dot_product

model = MatrixFactorizationModel(num_users, num_items)

 # Step 5: Compile the model
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.01),
loss='mse', metrics=['mae'])

 # Step 6: Train the model history = model.fit(train_dataset,
validation_data=test_dataset, epochs=10)

 # Step 7: Evaluate the model test_loss, test_mae =
model.evaluate(test_dataset) print(f"Test Loss:
{test_loss:.4f}, Test MAE: {test_mae:.4f}")

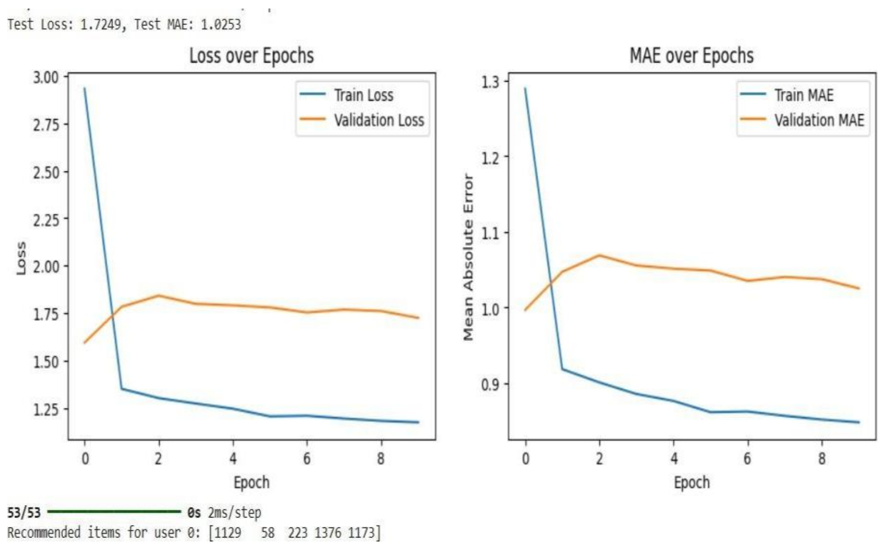
 # Step 8: Plot the training history
plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation
Loss') plt.xlabel('Epoch') plt.ylabel('Loss') plt.legend()
plt.title('Loss over Epochs')

plt.subplot(1, 2, 2)

```
plt.plot(history.history['mae'],       label='Train       MAE')
plt.plot(history.history['val_mae'],          label='Validation
MAE') plt.xlabel('Epoch')
plt.ylabel('Mean Absolute Error')
plt.legend()
plt.title('MAE over Epochs')


plt.show()


# Step 9: Make recommendations def
recommend(user_id, top_k=5):
    user_vector = tf.constant([user_id] * num_items, dtype=tf.int32)
    item_vector = tf.constant(list(range(num_items)), dtype=tf.int32)
    predictions    =    model.predict((user_vector,     item_vector))
    top_items = np.argsort(-predictions)[:top_k]
    return top_items


user_id = 0 # Example user recommended_items
= recommend(user_id)
print(f"Recommended items for user {user_id}: {recommended_items}"
```

Output :



Practical 5

Aim : Implementing a computer vision project, such as object detection or image segmentation.

Code :

```
!pip install torch torchvision numpy matplotlib opencv-python ultralytics


import cv2 import numpy as
np import matplotlib.pyplot as
plt
from ultralytics import YOLO # YOLOv5 library from ultralytics


# Step 1: Load the YOLOv5 Model print("Loading
YOLOv5 model...")
model = YOLO("yolov5s.pt") # Use the small version of YOLOv5 pre-trained on COCO dataset


# Step 2: Load an Image for Object Detection
image_path = "/example.jpg" # Replace with your image file path image
= cv2.imread(image_path)
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)


# Step 3: Perform Object Detection
print("Performing object detection...")
results = model.predict(image_rgb)


# Step 4: Visualize Results annotated_image = results[0].plot() # Annotated image with
bounding boxes and labels plt.figure(figsize=(10, 10))
plt.imshow(cv2.cvtColor(annotated_image,                    cv2.COLOR_BGR2RGB))
plt.axis("off")
```

```
plt.title("Object        Detection        Results")
plt.show()

# Step 5: Save the Annotated Image
output_path          =          "output.jpg"
cv2.imwrite(output_path,
annotated_image)
print(f"Annotated image saved to: {output_path}")

# Step 6: Print Detected Objects
print("Detected objects:") for box in
results[0].boxes.data.tolist():
   x1, y1, x2, y2, conf, cls = box
   print(f"Class: {results[0].names[int(cls)]}, Confidence: {conf:.2f}, Coordinates: ({x1:.2f},
{y1:.2f}), ({x2:.2f}, {y2:.2f})")
```

Output :


```
Annotated image saved to: output.jpg
Detected objects:
```

Practical 6

Aim : Training a generative adversarial network (GAN) for generating realistic images

Code :

```
!pip install torch torchvision matplotlib numpy

import torch import
torch.nn as nn import
torch.optim as optim
from torchvision import datasets, transforms
from torch.utils.data import DataLoader
import matplotlib.pyplot as plt

# Step 1: Define Generator and Discriminator class
Generator(nn.Module):
  def _init_(self, noise_dim, img_dim):
    super(Generator, self)._init_()
    self.model = nn.Sequential(
    nn.Linear(noise_dim, 128),
    nn.ReLU(), nn.Linear(128,
    256), nn.ReLU(),
    nn.Linear(256, 512),
    nn.ReLU(), nn.Linear(512,
    img_dim), nn.Tanh(),
    )
  def forward(self, x):
    return self.model(x)
```

```
class Discriminator(nn.Module):
    def _init_(self, img_dim):
        super(Discriminator, self)._init_()
        self.model = nn.Sequential(
        nn.Linear(img_dim, 512),
        nn.LeakyReLU(0.2),
        nn.Linear(512, 256),
        nn.LeakyReLU(0.2),
        nn.Linear(256, 1), nn.Sigmoid(),
        )

    def forward(self, x):
        return self.model(x)
```

 # Step 2: Define Constants and Hyperparameters

```
device = "cuda" if torch.cuda.is_available() else "cpu"
img_size = 28
img_dim = img_size * img_size
noise_dim = 100 batch_size =
64 epochs = 50
lr = 0.0002
```

 # Step 3: Prepare the MNIST Dataset

```
transform = transforms.Compose([transforms.ToTensor(), transforms.Normalize((0.5,),
(0.5,))])
dataset = datasets.MNIST(root="data", train=True, transform=transform, download=True)
dataloader = DataLoader(dataset, batch_size=batch_size, shuffle=True)
```

 # Step 4: Initialize Models, Loss, and Optimizers

```
generator = Generator(noise_dim,
```

img_dim).to(device) discriminator =
Discriminator(img_dim).to(device)

criterion = nn.BCELoss()
optimizer_g = optim.Adam(generator.parameters(), lr=lr) optimizer_d
= optim.Adam(discriminator.parameters(), lr=lr)

 # Step 5: Training Loop for
epoch in range(epochs):
for real_images, _ in dataloader: real_images =
real_images.view(-1, img_dim).to(device) batch_size =
real_images.size(0)

 # Labels for real and fake images real_labels =
torch.ones(batch_size, 1).to(device) fake_labels =
torch.zeros(batch_size, 1).to(device)

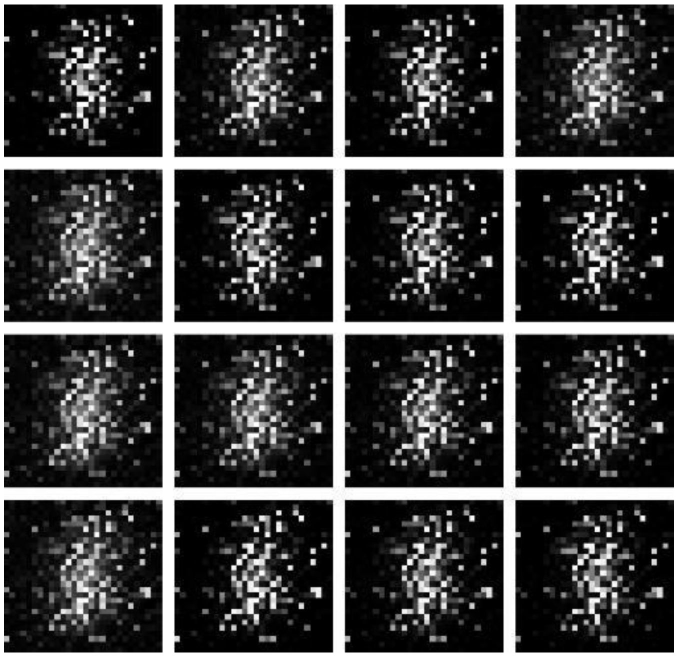
 # Train Discriminator
noise = torch.randn(batch_size, noise_dim).to(device)
fake_images = generator(noise) real_preds =
discriminator(real_images) fake_preds =
discriminator(fake_images.detach()) loss_d_real =
criterion(real_preds, real_labels) loss_d_fake =
criterion(fake_preds, fake_labels) loss_d =
(loss_d_real + loss_d_fake) / 2
optimizer_d.zero_grad() loss_d.backward()
optimizer_d.step()

 # Train Generator noise = torch.randn(batch_size,
noise_dim).to(device) fake_images =
generator(noise) fake_preds =

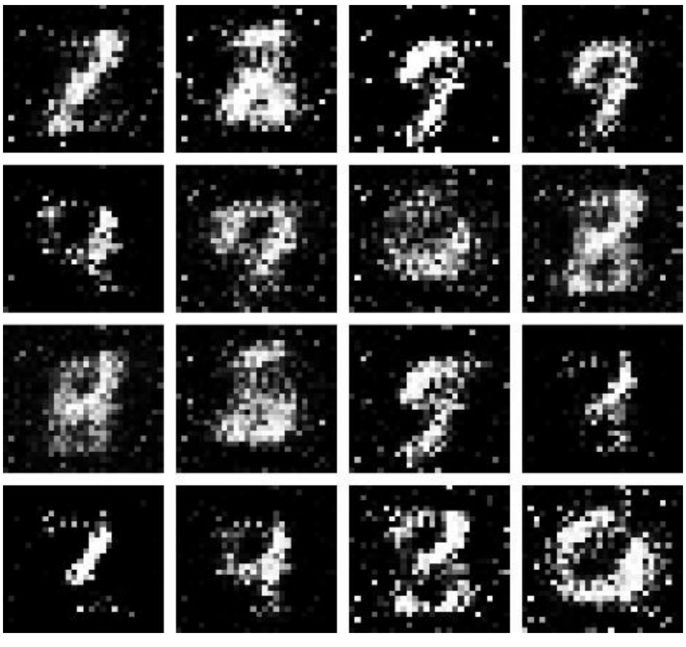
```
    discriminator(fake_images)             loss_g             =
    criterion(fake_preds,                              real_labels)
    optimizer_g.zero_grad() loss_g.backward()
    optimizer_g.step()

  # Print progress
  print(f"Epoch [{epoch+1}/{epochs}] | Loss D: {loss_d:.4f} | Loss G: {loss_g:.4f}")

  # Save generated samples every 10 epochs if
  (epoch + 1) % 10 == 0:
    noise = torch.randn(16, noise_dim).to(device)
    generated_images = generator(noise).view(-1, 1, img_size, img_size).cpu().detach()
    plt.figure(figsize=(8, 8)) for i in range(16):
      plt.subplot(4, 4, i + 1)
      plt.imshow(generated_images[i].squeeze(),                      cmap="gray")
      plt.axis("off")
    plt.tight_layout()
    plt.savefig(f"generated_images_epoch_{epoch+1}.png")
    plt.close()

# Step 6: Save the Generator Model
torch.save(generator.state_dict(),
"gan_generator.pth") print("Generator model saved as
gan_generator.pth")
```

Output :

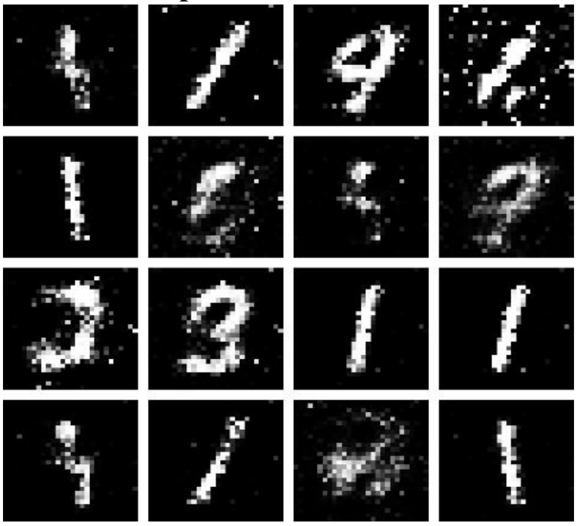
Generate after 10 Epoch



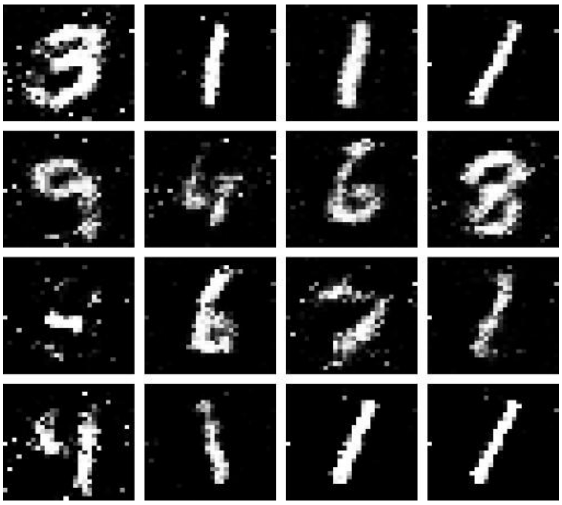
Generate after 20 Epoch



Generate 30 Epoch



Generate after 40 Epoch



Generate after 50 Epoch

Practical 7

Aim : Applying reinforcement learning algorithms to solve complex decision-making problems.

Code :

Step 1: Install Required Libraries

Run the following commands in your terminal to install the necessary libraries:

```
!pip install numpy gym matplotlib
```

Step 2: Python Code for Q-Learning

Save the following code as q_learning_cartpole.py.

```
import  gym  import
numpy as np
import matplotlib.pyplot as plt

# Step 1: Initialize Environment and Parameters env
= gym.make("CartPole-v1")
n_actions = env.action_space.n # Number of actions (2: left, right)
n_states = 20 # Discretize continuous state space episodes = 500 #
Number of episodes learning_rate = 0.1 # Learning rate (alpha)
discount_factor = 0.99 # Discount factor (gamma) epsilon = 1.0 #
Exploration rate epsilon_decay = 0.995 # Decay factor for epsilon
min_epsilon = 0.01 # Minimum epsilon

# Step 2: Helper Functions for Discretizing States
def discretize_state(state, state_bins):
```

```
    return tuple(np.digitize(state[i], state_bins[i]) for i in range(len(state)))

def create_bins(n_states, env):
    state_bins = [] for i in range(env.observation_space.shape[0]): low, high =
    env.observation_space.low[i],     env.observation_space.high[i]     bins     =
    np.linspace(low, high, n_states - 1) state_bins.append(bins)
    return state_bins

# Step 3: Initialize Q-Table and State Bins state_bins
= create_bins(n_states, env)
q_table = np.zeros((n_states,) * len(env.observation_space.shape) + (n_actions,))

#   Step   4:   Training   Loop
rewards  =  []  for  episode  in
range(episodes):
    state = discretize_state(env.reset()[0], state_bins) total_reward
    = 0
    done = False

    while not done:
        # Epsilon-greedy action selection if
        np.random.rand() < epsilon:
            action = np.random.choice(n_actions) # Explore else:
            action = np.argmax(q_table[state]) # Exploit

        # Take action and observe results next_state_raw,
        reward,  done,  _,  _  =  env.step(action)  next_state  =
        discretize_state(next_state_raw,                state_bins)
        total_reward += reward
```

 # Q-Learning update q_table[state][action] += learning_rate * (reward + discount_factor * np.max(q_table[next_state]) - q_table[state][action]
)

state = next_state

 # Decay epsilon

epsilon = max(min_epsilon, epsilon * epsilon_decay)

rewards.append(total_reward) print(f"Episode {episode + 1}/{episodes}, Reward: {total_reward}, Epsilon:
{epsilon:.3f}")

 # Step 5: Plot Rewards plt.plot(rewards)

plt.title("Total Rewards Over Episodes")

plt.xlabel("Episode") plt.ylabel("Total
Reward") plt.show()

 # Step 6: Save the Q-Table

np.save("q_table.npy", q_table)

print("Q-Table saved as 'q_table.npy'.")

 # Step 7: Test the Trained Model

```
state = discretize_state(env.reset()[0], state_bins) done
= False
total_reward = 0 while
not done:
    action = np.argmax(q_table[state]) # Use trained Q-Table
    next_state_raw, reward, done, _, _ = env.step(action) state
    = discretize_state(next_state_raw, state_bins)
    total_reward += reward
    env.render()

env.close() print(f"Total reward during test:
{total_reward}")
```

Practical 8

Aim : Utilizing transfer learning to improve model performance on limited datasets.

Code :

Step 1: Install Required Libraries

Run the following command to install the necessary libraries:

```
!pip install torch torchvision matplotlib numpy
```

Step 2: Python Code for Transfer Learning

Save the following code as transfer_learning.py.

```
import torch import
torch.nn as nn import
torch.optim as optim
from torchvision import datasets, models, transforms from
torch.utils.data import DataLoader
import matplotlib.pyplot as plt

# Step 1: Set Device and Hyperparameters device =
torch.device("cuda" if torch.cuda.is_available() else "cpu")
num_classes = 10 # CIFAR-10 has 10 classes batch_size = 32
epochs = 10
learning_rate = 0.001

# Step 2: Define Data Transformations transform
= transforms.Compose([ transforms.Resize((224,
```

224)), # Resize to match ResNet input size

transforms.ToTensor(),

transforms.Normalize(mean=[0.5, 0.5, 0.5], std=[0.5, 0.5, 0.5]),

])

Step 3: Load CIFAR-10 Dataset train_dataset = datasets.CIFAR10(root="data", train=True, transform=transform, download=True) test_dataset = datasets.CIFAR10(root="data", train=False, transform=transform, download=True)

train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True) test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=False)

 # Step 4: Load Pre-trained ResNet18 Model

model = models.resnet18(pretrained=True) for

param in model.parameters():

param.requires_grad = False # Freeze all layers

 # Replace the final fully connected layer for CIFAR-10

model.fc = nn.Linear(model.fc.in_features, num_classes)

model = model.to(device)

 # Step 5: Define Loss and Optimizer criterion

= nn.CrossEntropyLoss()

optimizer = optim.Adam(model.fc.parameters(), lr=learning_rate)

 # Step 6: Training Loop def train(model,

loader, criterion, optimizer):

model.train()

running_loss = 0.0 correct

```
= 0 total = 0 for inputs,
labels in loader:
    inputs, labels = inputs.to(device), labels.to(device)

    optimizer.zero_grad() outputs
    = model(inputs) loss =
    criterion(outputs, labels)
    loss.backward()
    optimizer.step()

    running_loss  +=  loss.item()
    _,            predicted          =
    outputs.max(1)     total     +=
    labels.size(0)
    correct += predicted.eq(labels).sum().item()

accuracy = 100. * correct / total
return running_loss / len(loader), accuracy
```

 # Step 7: Testing Loop def

```
test(model, loader, criterion):
    model.eval()
    running_loss = 0.0
    correct = 0 total = 0
    with
    torch.no_grad():
        for inputs, labels in loader:
            inputs, labels = inputs.to(device), labels.to(device)
```

```
        outputs = model(inputs) loss =
        criterion(outputs, labels)
        running_loss += loss.item()

        _, predicted = outputs.max(1) total
        += labels.size(0)
        correct += predicted.eq(labels).sum().item()

    accuracy = 100. * correct / total
    return running_loss / len(loader), accuracy
```

 # Step 8: Train and Evaluate

```
train_losses, test_losses = [], []
train_accuracies, test_accuracies = [], []

for epoch in range(epochs):
    train_loss, train_acc = train(model, train_loader, criterion, optimizer) test_loss,
    test_acc = test(model, test_loader, criterion)

    train_losses.append(train_loss)
    test_losses.append(test_loss)
    train_accuracies.append(train_acc)
    test_accuracies.append(test_acc)

    print(f"Epoch {epoch+1}/{epochs}")
    print(f"Train Loss: {train_loss:.4f}, Train Accuracy: {train_acc:.2f}%") print(f"Test
    Loss: {test_loss:.4f}, Test Accuracy: {test_acc:.2f}%")
```

 # Step 9: Plot Training and Testing Metrics

```
plt.figure(figsize=(12, 5)) plt.subplot(1, 2, 1)
```

plt.plot(train_losses, label="Train Loss") plt.plot(test_losses, label="Test Loss") plt.legend() plt.title("Loss")

plt.subplot(1, 2, 2)
plt.plot(train_accuracies, label="Train Accuracy")
plt.plot(test_accuracies, label="Test Accuracy")
plt.legend() plt.title("Accuracy") plt.show()

 # Step 10: Save the Model
torch.save(model.state_dict(),
"resnet18_cifar10.pth") print("Model saved as 'resnet18_cifar10.pth'.")

Practical 9

Aim : Building a deep learning model for time series forecasting or anomaly detection

Code :

```
!pip install numpy pandas matplotlib scikit-learn tensorflow

import numpy as np import
pandas as pd import
matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense

# Step 1: Load the Dataset url =
"https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-passengers.csv" data
= pd.read_csv(url, usecols=[1], header=0)
data = data.values.astype("float32") # Ensure the data is float

# Step 2: Visualize the Data plt.plot(data)
plt.title("Airline Passengers Over Time")
plt.xlabel("Time")
plt.ylabel("Passengers")
plt.show()

# Step 3: Normalize the Data scaler =
MinMaxScaler(feature_range=(0, 1))
scaled_data = scaler.fit_transform(data)
```

 # Step 4: Prepare the Data for LSTM

```
def create_dataset(dataset, look_back=1):
X, y = [], [] for i in range(len(dataset) -
look_back):
        X.append(dataset[i:(i + look_back), 0])
        y.append(dataset[i + look_back, 0])
    return np.array(X), np.array(y)

look_back = 12 # Use 12 months (1 year) as input to predict the next value
X, y = create_dataset(scaled_data, look_back)
X = X.reshape((X.shape[0], X.shape[1], 1)) # Reshape for LSTM [samples, time_steps, features]
```

 # Step 5: Split Data into Training and Testing Sets train_size

```
= int(len(X) * 0.8)
X_train, X_test = X[:train_size], X[train_size:] y_train,
y_test = y[:train_size], y[train_size:]
```

 # Step 6: Build the LSTM Model model

```
= Sequential([
    LSTM(50, activation="relu", input_shape=(look_back, 1)), Dense(1)
])
model.compile(optimizer="adam", loss="mean_squared_error")
```

 # Step 7: Train the Model history = model.fit(X_train, y_train, epochs=50, batch_size=32, validation_data=(X_test, y_test), verbose=1)

 # Step 8: Evaluate the Model

```
loss = model.evaluate(X_test, y_test, verbose=0) print(f"Test
Loss: {loss:.4f}")
```

 # Step 9: Predict and Inverse Transform

y_pred = model.predict(X_test) y_pred =

scaler.inverse_transform(y_pred)

y_test_actual = scaler.inverse_transform(y_test.reshape(-1, 1))

 # Step 10: Plot Actual vs Predicted

plt.figure(figsize=(10, 5)) plt.plot(y_test_actual,

label="Actual") plt.plot(y_pred,

label="Predicted") plt.title("Actual vs Predicted

Airline Passengers") plt.xlabel("Time")

plt.ylabel("Passengers") plt.legend()

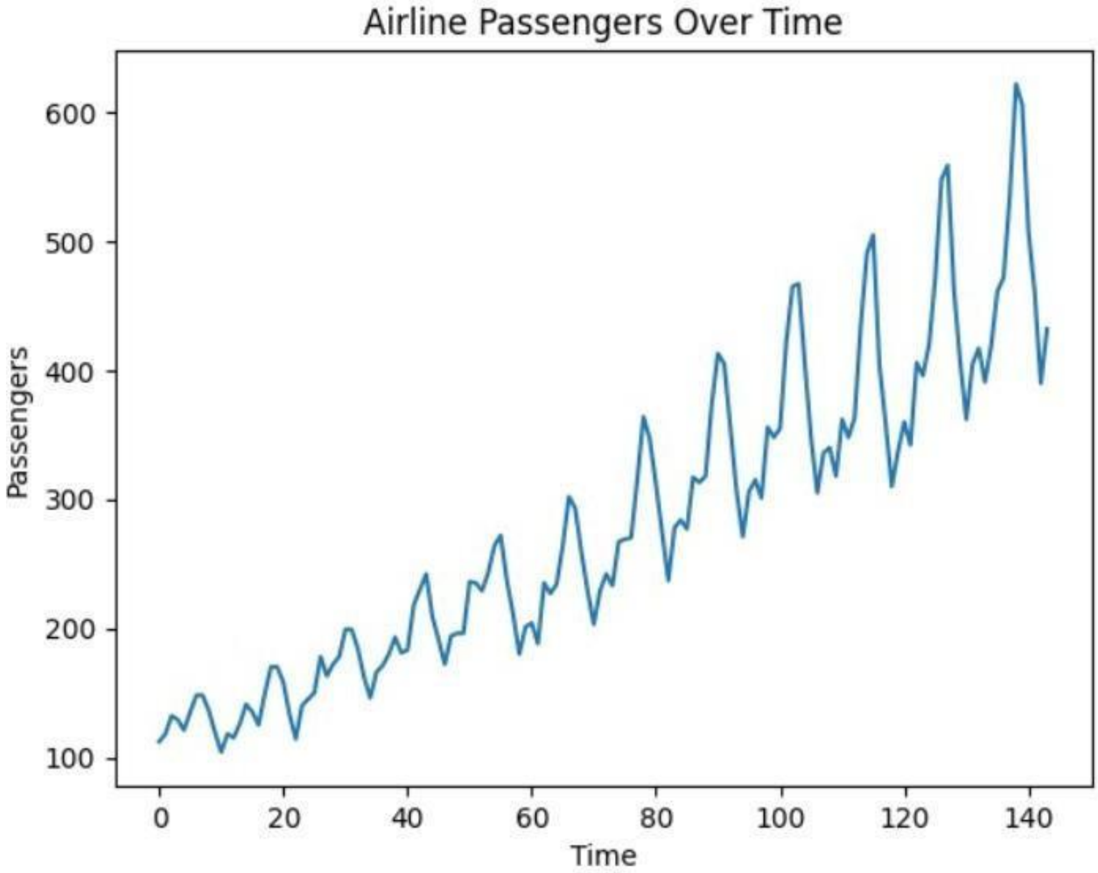
plt.show()

 # Step 11: Save the Model

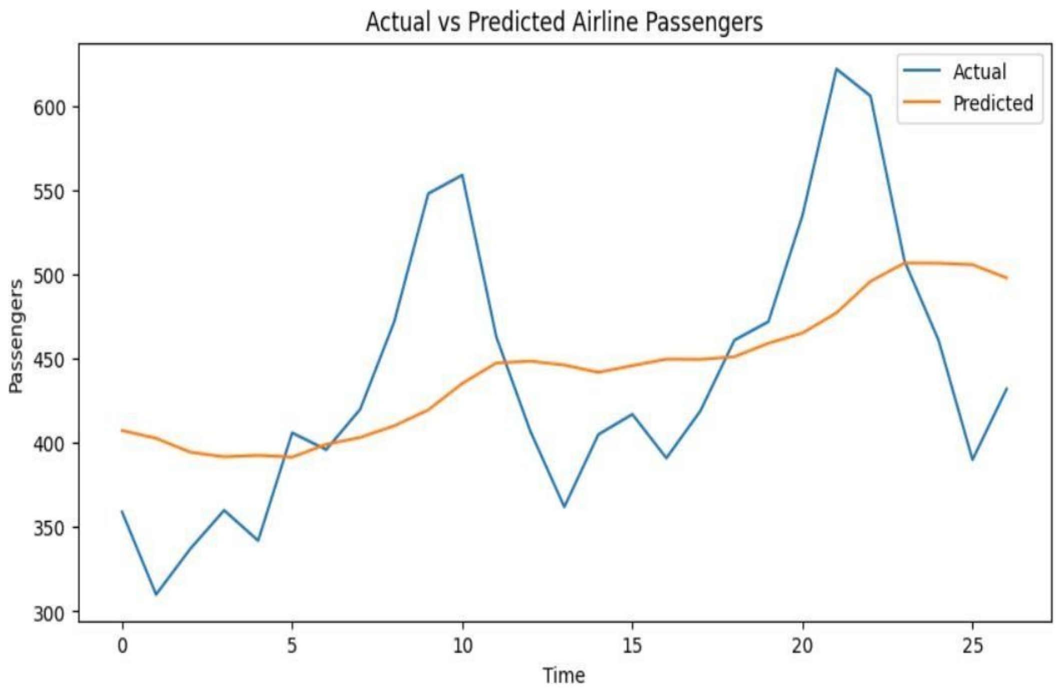
model.save("lstm_time_series.h5")

print("Model saved as 'lstm_time_series.h5'.")

Output :



Airline Passengers Over Time
Passengers
Time
600
500
400
300
200
100
0
20
40
60
80
100
120
140



Actual vs Predicted Airline Passengers
Actual
Predicted
Passengers
Time
600
550
500
450
400
350
300
0
5
10
15
20
25

Practical 10

Aim : Implementing a machine learning pipeline for automated feature engineering and model selection.

Code :

Step 1: Install Required Libraries

Run the following command to install the required libraries

pip install pandas numpy scikit-learn

Step 2: Python Code for Machine Learning Pipeline

Save the following code as ml_pipeline.py.

```
import pandas as pd import numpy as np from
sklearn.model_selection import train_test_split, GridSearchCV from
sklearn.pipeline import Pipeline from sklearn.compose import
ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.feature_selection import SelectKBest, f_classif from
sklearn.ensemble import RandomForestClassifier from sklearn.svm
import SVC
from sklearn.metrics import classification_report, accuracy_score
```

 # Step 1: Load Dataset

```
# Using the Titanic dataset for demonstration
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv" data
= pd.read_csv(url)
```

Step 2: Basic Preprocessing

Drop unnecessary columns

data = data.drop(["PassengerId", "Name", "Ticket", "Cabin"], axis=1)

# **Handle missing values** data["Age"].fillna(data["Age"].median(),
inplace=True)
data["Embarked"].fillna(data["Embarked"].mode()[0],
inplace=True)

Separate features and target X
= data.drop("Survived", axis=1) y
= data["Survived"]

Step 3: Split Data into Train and Test Sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 4: Define Preprocessing Steps

Numerical features: Scale values
numerical_features = ["Age", "Fare"]
numerical_transformer =
Pipeline(steps=[("scaler",
StandardScaler())
])

# Categorical features: One-hot encode
categorical_features = ["Sex", "Embarked", "Pclass"]
categorical_transformer = Pipeline(steps=[
  ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

```
# Combine preprocessors into a column transformer
preprocessor = ColumnTransformer(
transformers=[
        ("num", numerical_transformer, numerical_features),
        ("cat", categorical_transformer, categorical_features)
    ]
)

# Step 5: Define Feature Selection and Model Options feature_selection
= SelectKBest(score_func=f_classif)


# Define candidate models models
= {
    "RandomForest": RandomForestClassifier(random_state=42),
    "SVC": SVC(probability=True, random_state=42)
}

# Step 6: Create the Pipeline pipeline
= Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("feature_selection", feature_selection),
    ("classifier", RandomForestClassifier())
])

# Step 7: Define Grid Search for Hyperparameter Tuning
param_grid = {
    "feature_selection_k": [5, 6, 7],
    "classifier": [models["RandomForest"], models["SVC"]],
    "classifier__n_estimators": [100, 200] if "n_estimators" in
RandomForestClassifier().get_params() else [None],
```

"classifier_C": [0.1, 1, 10] if "C" in SVC().get_params() else [None]

}

grid_search = GridSearchCV(pipeline, param_grid, cv=3, scoring="accuracy", verbose=2)

 # Step 8: Train the Model

grid_search.fit(X_train, y_train)

 # Step 9: Evaluate the Model best_model = grid_search.best_estimator_ y_pred = best_model.predict(X_test)

print("Best Parameters:", grid_search.best_params_) print("Accuracy on Test Set:", accuracy_score(y_test, y_pred)) print("\nClassification Report:\n", classification_report(y_test, y_pred))

Practical 11

Aim : Using advanced optimization techniques like evolutionary algorithms or Bayesian optimization for hyperparameter tuning.

Code :

```
!pip install numpy pandas scikit-learn scikit-optimize


import numpy as np import
pandas as pd
from sklearn.datasets import load_iris from sklearn.model_selection
import train_test_split, cross_val_score from sklearn.ensemble import
RandomForestClassifier from skopt import BayesSearchCV
from sklearn.metrics import accuracy_score, classification_report


# Step 1: Load the Dataset data
= load_iris()
X, y = data.data, data.target


# Step 2: Split the Data into Training and Testing Sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


# Step 3: Define the Model
model = RandomForestClassifier(random_state=42)


# Step 4: Define the Search Space for Hyperparameters
param_space = {
    "n_estimators": (10, 200),        # Number of trees in the forest
    "max_depth": (1, 20),             # Maximum depth of each tree
    "min_samples_split": (2, 10),   # Minimum samples to split a node
    "min_samples_leaf": (1, 10),       # Minimum samples at each leaf
```

```
    "max_features": ["sqrt", "log2", None] # Number of features considered for split
}

# Step 5: Use Bayesian Optimization for Hyperparameter Tuning
optimizer = BayesSearchCV( estimator=model,
search_spaces=param_space, n_iter=30, # Number of iterations to
search cv=3,      # 3-fold cross-validation random_state=42, n_jobs=-
1
)

# Step 6: Train the Optimized Model
print("Starting Bayesian Optimization...")
optimizer.fit(X_train, y_train)

# Step 7: Evaluate the Best Model
best_model = optimizer.best_estimator_
y_pred = best_model.predict(X_test)

print("\nBest Parameters:", optimizer.best_params_) print("Accuracy
on Test Set:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# Optional: Save the Best Model import joblib
joblib.dump(best_model,
"optimized_rf_model.pkl") print("\nModel saved as
'optimized_rf_model.pkl'.")
```

Output :

```
Best Parameters: OrderedDict([('max_depth', 16), ('max_features', 'log2'), ('min_samples_leaf', 6), ('min_samples_split', 8), ('n_estimators', 182)])
Accuracy on Test Set: 1.0

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30


Model saved as 'optimized_rf_model.pkl'.
```

Practical 12

Aim : Deploying a machine learning model in a production environment using containerization and cloud services

Code :

Step 1: Install Required Libraries

```
!pip install scikit-learn pandas fastapi uvicorn
```

Step 2: Build the Machine Learning Model

Save the following Python script as train_model.py. This script trains the model and saves it.

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.ensemble import RandomForestClassifier import
joblib

# Step 1: Load Dataset data
= load_iris()
X, y = data.data, data.target

# Step 2: Train Model
model = RandomForestClassifier(random_state=42) model.fit(X,
y)

#    Step    3:    Save    Model
joblib.dump(model,    "iris_model.pkl")
print("Model saved as iris_model.pkl")
```

Run the script to save the trained model:

bash
Copy code python

train_model.py

Step 3: Create the API

Save the following Python script as app.py.

python Copy code from fastapi

import FastAPI from pydantic

import BaseModel import joblib

import numpy as np

 # Step 1: Load the trained model

model = joblib.load("iris_model.pkl")

 # Step 2: Initialize FastAPI

app = FastAPI()

 # Step 3: Define input schema class

IrisRequest(BaseModel):

    sepal_length: float

    sepal_width: float

    petal_length: float

    petal_width: float

 # Step 4: Define prediction endpoint

@app.post("/predict/") def

predict(iris: IrisRequest):

   features = np.array([[iris.sepal_length, iris.sepal_width, iris.petal_length,
iris.petal_width]])

    prediction = model.predict(features)

    species = ["setosa", "versicolor", "virginica"]

return {"prediction": species[prediction[0]]}

Run the API locally for testing: bash Copy code

uvicorn app:app --host 0.0.0.0 --port 8000

Test the API in your browser or with a tool like **Postman**:

- URL: http://127.0.0.1:8000/predict/ Example Request Body:

json

Copy code

{
    "sepal_length": 5.1,
    "sepal_width": 3.5,
    "petal_length": 1.4,
    "petal_width": 0.2 }

Step 4: Create a Dockerfile Save the following as Dockerfile.

dockerfile

Copy code

```
# Use an official Python runtime as the base image
FROM python:3.9-slim

# Set the working directory in the container
WORKDIR /app

# Copy the current directory contents into the container
COPY . /app
# Install required Python libraries
RUN pip install --no-cache-dir fastapi uvicorn scikit-learn joblib
```

Expose the API port

EXPOSE 8000

Command to run the application

CMD ["uvicorn", "app:app", "--host", "0.0.0.0", "--port", "8000"]

Step 5: Build and Run the Docker Container

1. **Build the Docker image**:

bash Copy

code

docker build -t iris-api .

2. **Run the Docker container**:

bash Copy code docker run -d -p

8000:8000 iris-api

3. **Test the API**:
   - URL: http://localhost:8000/predict/
   - Use the same JSON request as earlier.

Step 6: Deploy to a Cloud Service (Optional)

1. **Prepare the Docker Image**:
   - Tag the image for a container registry (e.g., Docker Hub, AWS ECR, or GCP Artifact Registry):

bash Copy

code

docker tag iris-api <your_dockerhub_username>/iris-api docker

push <your_dockerhub_username>/iris-api

2. **Deploy to AWS ECS (Example)**:
   - Create an ECS cluster. ○ Use the Docker image in a task definition.
   - Deploy the task to the cluster.

3. **Other Options**:
    - Use **AWS Lambda** with **API Gateway**.
    - Deploy on **Google Cloud Run** or **Azure App Service** for managed hosting.

Step 7: Dataset

- **Iris Dataset**:

    ○ Included with sklearn.datasets for demonstration purposes. ○ Automatically loaded in the train_model.py script.

Practical 13

Aim : Use Python libraries such as GPT-2 or textgenrnn to train generative models on a corpus of text data and generate new text based on the patterns it has learned.

Code :

Step 1: Install Required Libraries

Run the following commands to install necessary libraries:

```
!pip install transformers datasets torch
```

Step 2: Prepare a Text Dataset

For demonstration, we'll use the Tiny Shakespeare Corpus available via Hugging Face's datasets library. Alternatively, you can use your own dataset.

Step 3: Python Code for Training and Generating Text

Save the following code as train_gpt2.py.

```
import os

from datasets import load_dataset

from transformers import GPT2LMHeadModel, GPT2Tokenizer, Trainer, TrainingArguments


# Step 1: Load the Dataset print("Loading

dataset...")

dataset = load_dataset("tiny_shakespeare")


# Split into train and test sets

train_data   =   dataset["train"]

test_data = dataset["test"]
```

 # Step 2: Load Pre-trained GPT-2 Tokenizer and Model

print("Loading GPT-2 tokenizer and model...")

tokenizer = GPT2Tokenizer.from_pretrained("gpt2")

model = GPT2LMHeadModel.from_pretrained("gpt2")

 # Step 3: Tokenize the Dataset def

tokenize_function(examples):

return tokenizer(examples["text"], truncation=True, padding="max_length", max_length=512)

print("Tokenizing dataset...")

tokenized_train = train_data.map(tokenize_function, batched=True) tokenized_test = test_data.map(tokenize_function, batched=True)

 # Step 4: Define Training Arguments

training_args = TrainingArguments(

output_dir="./results",

evaluation_strategy="epoch",

learning_rate=5e-5,

weight_decay=0.01,

per_device_train_batch_size=4,

num_train_epochs=3,

save_total_limit=2,

logging_dir="./logs",

logging_steps=10,

)

 # Step 5: Initialize Trainer trainer

= Trainer(

```
    model=model,
    args=training_args,
    train_dataset=tokenized_train,
    eval_dataset=tokenized_test,
)
```

 # Step 6: Train the Model

```
print("Starting  training...")
trainer.train()
```

 # Step 7: Save the Fine-Tuned Model

```
model.save_pretrained("./fine_tuned_gpt2")
tokenizer.save_pretrained("./fine_tuned_gpt2")
print("Model saved to './fine_tuned_gpt2'.")
```

 # Step 8: Generate Text Using the Fine-Tuned Model

```
print("Generating new text...") model.eval()

input_text = "To be or not to be, that is the" inputs =
tokenizer.encode(input_text, return_tensors="pt")
outputs = model.generate(inputs, max_length=100, num_return_sequences=1, temperature=0.7)

generated_text = tokenizer.decode(outputs[0], skip_special_tokens=True)
print("\nGenerated Text:\n") print(generated_text)
```

Practical 14

Aim : Experiment with neural networks like GANs (Generative Adversarial Networks) using Python libraries like TensorFlow or PyTorch to generate new images based on a dataset of images.

Code :

```
!pip install torch torchvision matplotlib


import torch import
torch.nn as nn import
torch.optim as optim
from torchvision import datasets, transforms
from torch.utils.data import DataLoader
import matplotlib.pyplot as plt
import os


# Step 1: Set Device Configuration
device = torch.device("cuda" if torch.cuda.is_available() else "cpu") print(f"Using
device: {device}")


# Step 2: Define Generator class
Generator(nn.Module):
  def _init_(self, noise_dim, img_dim):
    super(Generator, self)._init_()
    self.gen = nn.Sequential(
    nn.Linear(noise_dim, 256),
    nn.ReLU(), nn.Linear(256,
    512), nn.ReLU(),
      nn.Linear(512, img_dim),
      nn.Tanh()
    )
```

def forward(self, x):

return self.gen(x)

 # Step 3: Define Discriminator class

Discriminator(nn.Module):

def _init_(self, img_dim):

super(Discriminator, self)._init_()

self.disc = nn.Sequential(

nn.Linear(img_dim, 512),

nn.LeakyReLU(0.2),

nn.Linear(512, 256),

nn.LeakyReLU(0.2),

nn.Linear(256, 1), nn.Sigmoid()

)

def forward(self, x): return

self.disc(x)

 # Step 4: Define Hyperparameters noise_dim

= 100

img_dim = 28 * 28 # 28x28 images flattened

batch_size = 64 lr = 0.0002

epochs = 50

 # Step 5: Load Dataset

transform = transforms.Compose([transforms.ToTensor(), transforms.Normalize((0.5,), (0.5,))])

dataset = datasets.MNIST(root="data", train=True, transform=transform, download=True)

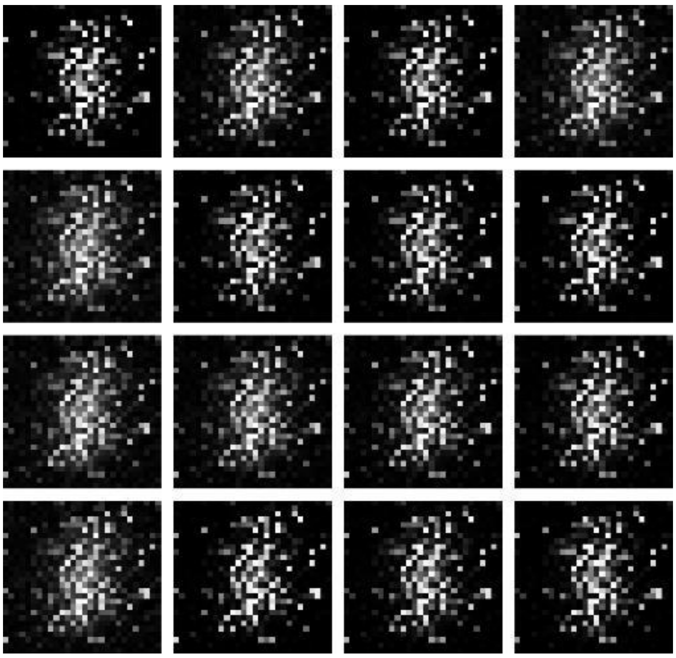
dataloader = DataLoader(dataset, batch_size=batch_size, shuffle=True)

```
# Step 6: Initialize Models, Optimizers, and Loss Function
gen = Generator(noise_dim, img_dim).to(device) disc =
Discriminator(img_dim).to(device) criterion = nn.BCELoss()
opt_gen = optim.Adam(gen.parameters(), lr=lr) opt_disc
= optim.Adam(disc.parameters(), lr=lr)

# Step 7: Training Loop
print("Starting Training...")
for epoch in range(epochs):
  for batch_idx, (real, _) in enumerate(dataloader):
    real = real.view(-1, img_dim).to(device) batch_size
    = real.size(0)

    # Train Discriminator
    noise = torch.randn(batch_size, noise_dim).to(device)
    fake = gen(noise) disc_real = disc(real).view(-1)
    disc_fake = disc(fake.detach()).view(-1)
    loss_disc = criterion(disc_real, torch.ones_like(disc_real)) + \ criterion(disc_fake,
          torch.zeros_like(disc_fake))
    opt_disc.zero_grad()
    loss_disc.backward()
    opt_disc.step()

    # Train Generator
    output = disc(fake).view(-1)
    loss_gen = criterion(output,
    torch.ones_like(output)) opt_gen.zero_grad()
    loss_gen.backward() opt_gen.step()

print(f"Epoch [{epoch+1}/{epochs}] | Loss D: {loss_disc:.4f}, Loss G: {loss_gen:.4f}")
```

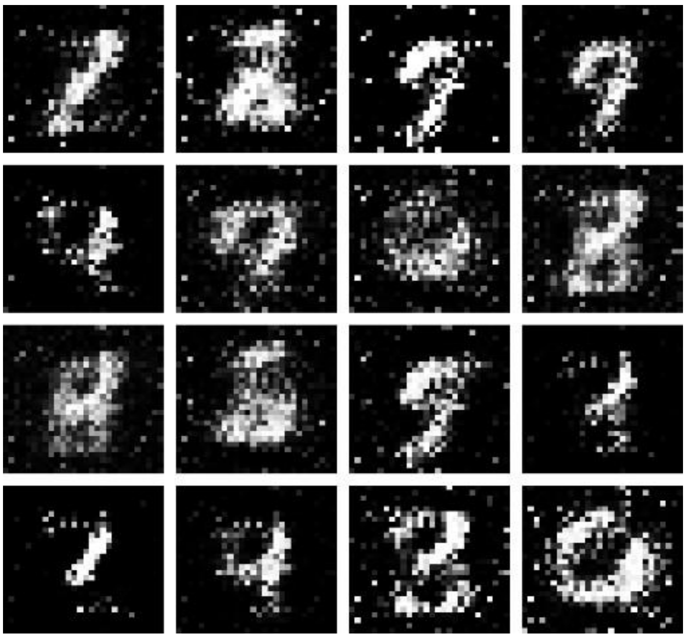
```
# Save and Display Sample Images if
(epoch + 1) % 10 == 0:
    with torch.no_grad():
        fake_images = gen(torch.randn(16, noise_dim).to(device)).view(-1, 1, 28, 28)
    plt.figure(figsize=(4, 4)) for
    i in range(16):
        plt.subplot(4,              4,              i+1)
        plt.imshow(fake_images[i][0].cpu(),
        cmap="gray") plt.axis("off")
    plt.tight_layout()
    os.makedirs("generated_images", exist_ok=True)
    plt.savefig(f"generated_images/epoch_{epoch+1}.png
    ") plt.close()

print("Training Complete. Generated images are saved in 'generated_images' folder.")
```

Output :

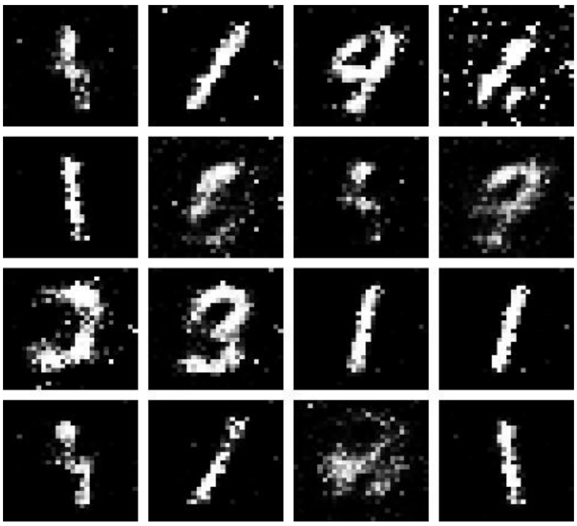
Generate after 10 Epoch



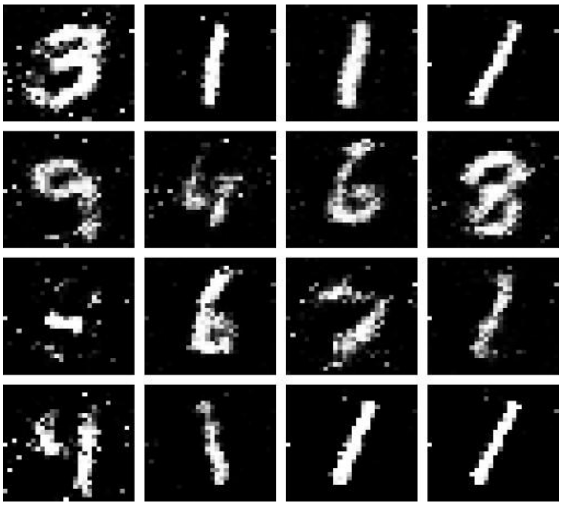
Generate after 20 Epoch



Generate 30 Epoch



Generate after 40 Epoch



Generate after 50 Epoch

PRACTICAL JOURNAL

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Submitted by

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Under the guidance of

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Department of Information Technology

Kalyan, Maharashtra

Academic Year 2025-26

The Kalyan Wholesale Merchants Education Society's

Laxman Devram Sonawane College, Kalyan (W) 421301

Department of Information Technology
Masters of Science – Part II

Certificate

This is to certify that **Mr. Tejas Anil Kapade**, Seat number______________, studying in Masters of Science in Information Technology Part II , Semester III has satisfactorily completed the practical of "Advanced Artificial Intelligence " as prescribed by University of Mumbai, during the academic year 2025-2026

Subject In-charge　　Coordinator In-charge　　External Examiner

College Seal

MACHINE LEARNING

INDEX

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Practical 1: Data Pre-processing and Exploration

1a. Load a CSV dataset. Handle missing values, inconsistent formatting, and outliers.

Code :

1. **Import Libraries**

 # Import necessary libraries import

pandas as pd import numpy as np import

seaborn as sns

import matplotlib.pyplot as plt

2. **Load the Dataset**

 # Load the Titanic dataset from a URL
url="https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv" data = pd.read_csv(url)

 # Display the first few rows

print(data.head())

3. **Handle Missing Values # Check for missing values**
   print("Missing values in each column:")
   print(data.isnull().sum())

 # Fill missing values in 'Age' with the mean data['Age'].fillna(data['Age'].mean(),

inplace=True)

 # Fill missing values in 'Embarked' with the most common value

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True) #

Drop rows where 'Cabin' is missing (too many NaNs)

data.drop(columns=['Cabin'], inplace=True)

 # Verify missing values are handled

print("\nAfter handling missing values:")

print(data.isnull().sum())

4. **Fix Inconsistent Formatting**

 # Fix inconsistent formatting in the 'Sex' column data['Sex']

= data['Sex'].str.lower().str.strip()

 # Verify unique values print("\nUnique values in 'Sex'

column after formatting:") print(data['Sex'].unique())

5. **Detect and Handle Outliers # Boxplot for the 'Fare' column**

   sns.boxplot(data['Fare'], color='skyblue') plt.title('Boxplot of Fare') plt.show()

 # Detect outliers using the IQR method

Q1 = data['Fare'].quantile(0.25)
Q3 = data['Fare'].quantile(0.75)
IQR = Q3 - Q1 lower_bound =
Q1 - 1.5 * IQR upper_bound =
Q3 + 1.5 * IQR #

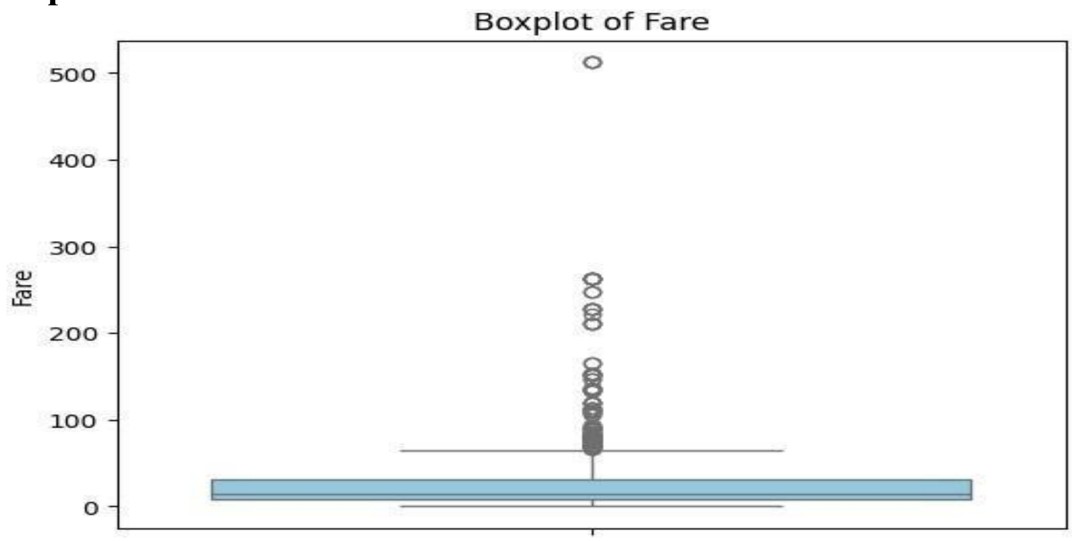
Capping outliers

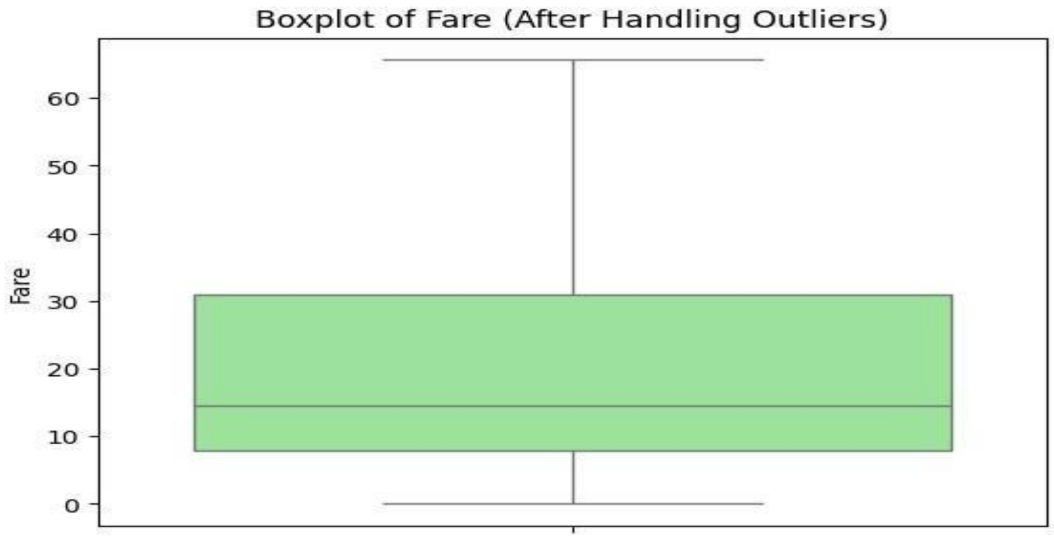
data['Fare'] = np.where(data['Fare'] > upper_bound, upper_bound, np.where(data['Fare'] < lower_bound, lower_bound, data['Fare'])) **# Verify with an updated boxplot**

sns.boxplot(data['Fare'], color='lightgreen') plt.title('Boxplot of Fare (After Handling Outliers)') plt.show()

6. **Save the Cleaned Dataset # Save the cleaned dataset**
   data.to_csv('cleaned_titanic.csv', index=False)
   print("\nCleaned dataset saved as 'cleaned_titanic.csv'") .

Output :





1b. Load a dataset, calculate descriptive summary statistics, create visualizations using different graphs, and identify potential features and target variables Note:

Explore Univariate and Bivariate graphs (Matplotlib) and Seaborn for visualization Code :

1. Import Necessary Libraries # Import required libraries

import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns

2. Load the Dataset

 # Load the dataset from the URL url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv" data

= pd.read_csv(url) # **Display the first few rows** print("First 5 rows of the dataset:") print(data.head())

3. Calculate Descriptive Summary Statistics # Dataset information print("\nDataset Info:") print(data.info())

Summary statistics for numerical columns print("\nDescriptive Statistics for Numerical Columns:") print(data.describe())

Check unique values for categorical columns print("\nUnique values in 'species' column:") print(data['species'].value_counts())

4. Univariate Analysis

Histograms for numerical columns

data.hist(figsize=(10,8), color='skyblue', edgecolor='black') plt.suptitle("Histograms of Numerical Features") plt.show()

# **Bar plot for 'species' column** sns.countplot(x='species', data=data, palette='pastel')
plt.title("Count of Each Species") plt.show()

5. Bivariate Analysis # Scatter plot for two features plt.figure(figsize=(8, 6))

plt.scatter(data['sepal_length'], data['sepal_width'], alpha=0.7, c='blue')

plt.title("Sepal Length vs Sepal Width") plt.xlabel("Sepal Length")

plt.ylabel("Sepal Width") plt.show()

Pairplot to visualize relationships between features sns.pairplot(data, hue='species', palette='husl', diag_kind='kde') plt.suptitle("Pairplot of Features by Species", y=1.02) plt.show()

Boxplot for petal_length across species sns.boxplot(x='species', y='petal_length', data=data, palette='Set3') plt.title("Boxplot of Petal Length by Species") plt.show()

6. Identify Potential Features and Target Variables # Separate features and target features = data.drop(columns=['species'])

 # Drop the target column

target = data['species'] #

Target variable

print("\nFeatures:")

print(features.head())

print("\nTarget:")

print(target.head())

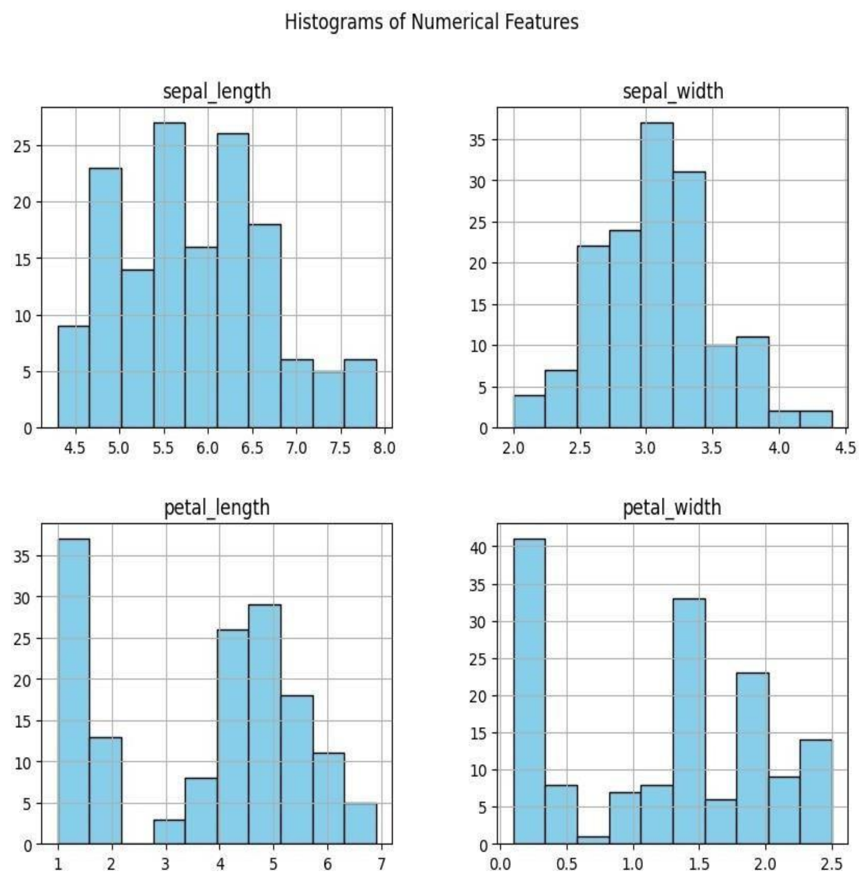
 # Visualize target distribution
sns.countplot(x=target,
palette='viridis') plt.title("Target
Variable Distribution") plt.show()

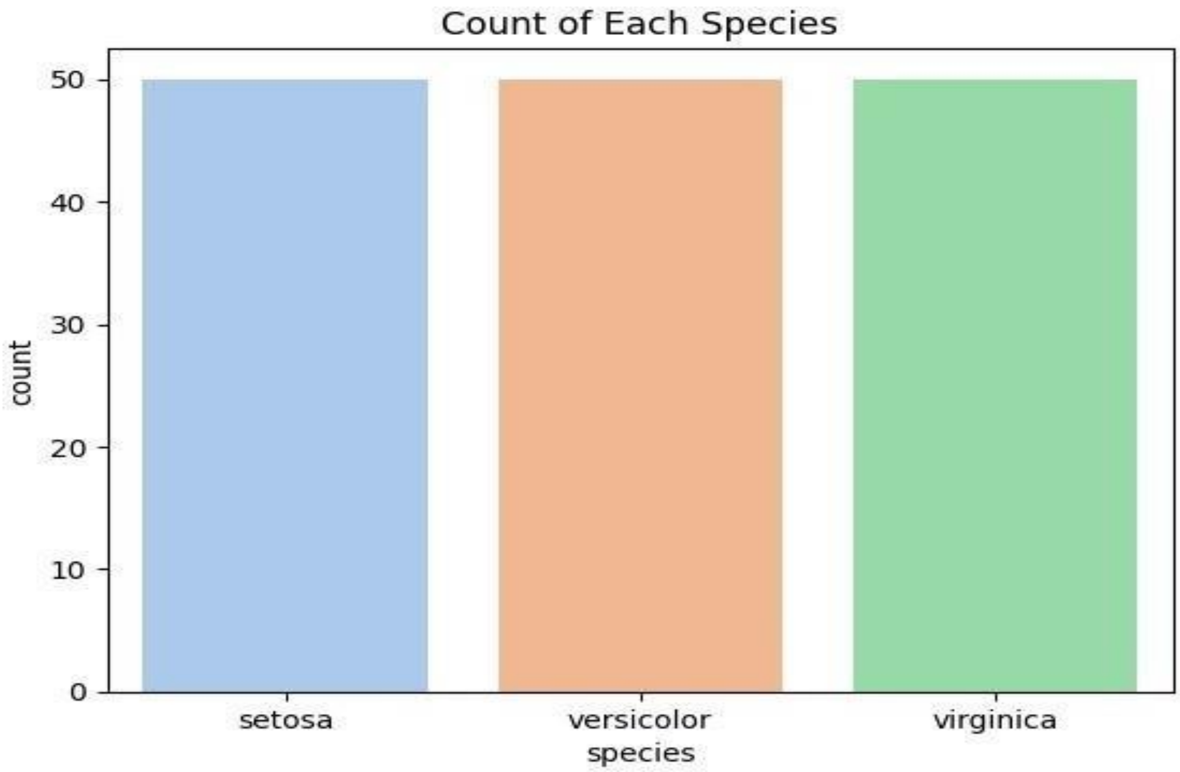
7. Save the Cleaned and Processed Dataset # Save the dataset

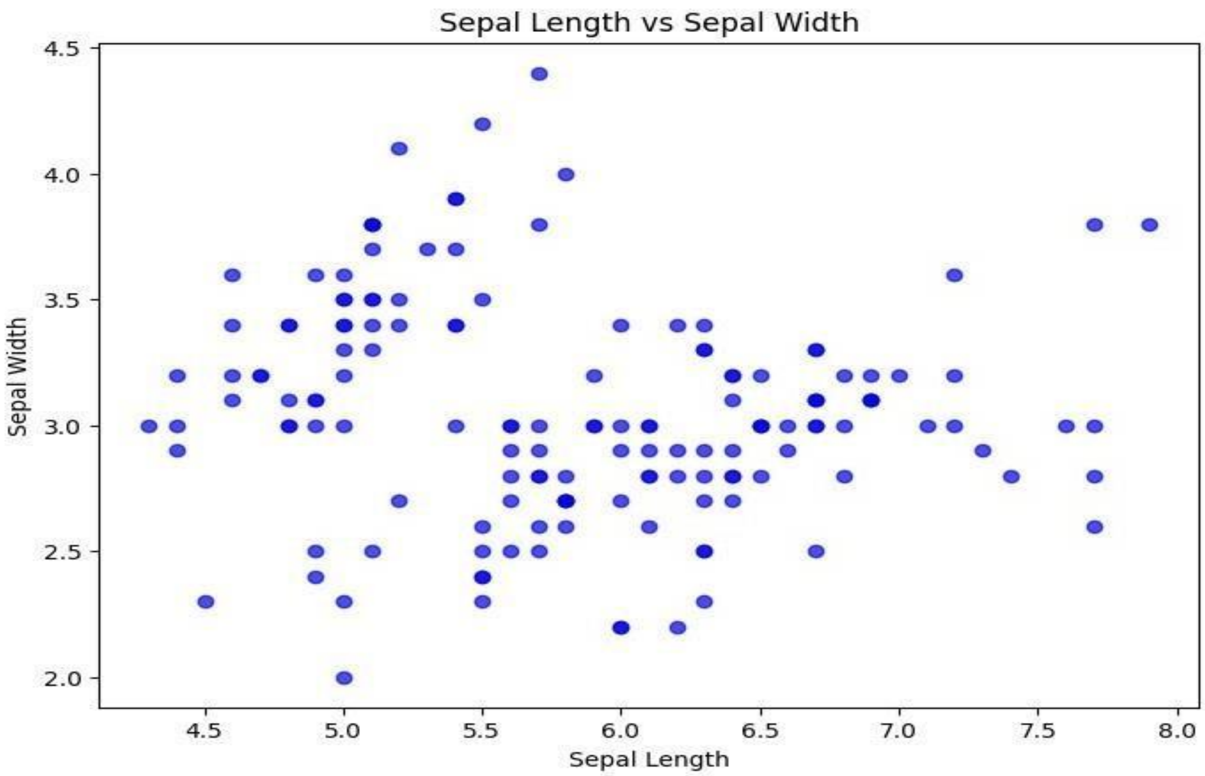
data.to_csv('processed_iris.csv', index=False)

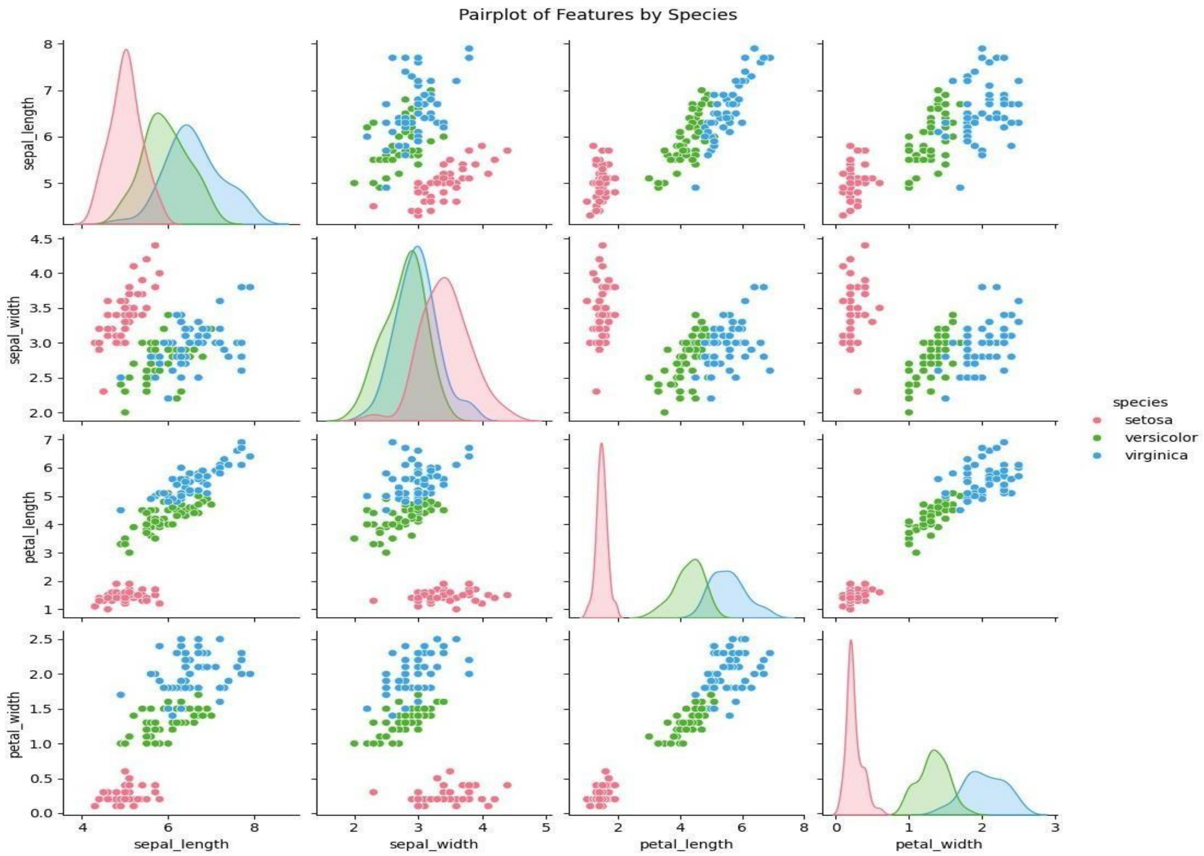
print("\nProcessed dataset saved as 'processed_iris.csv'")

Output :

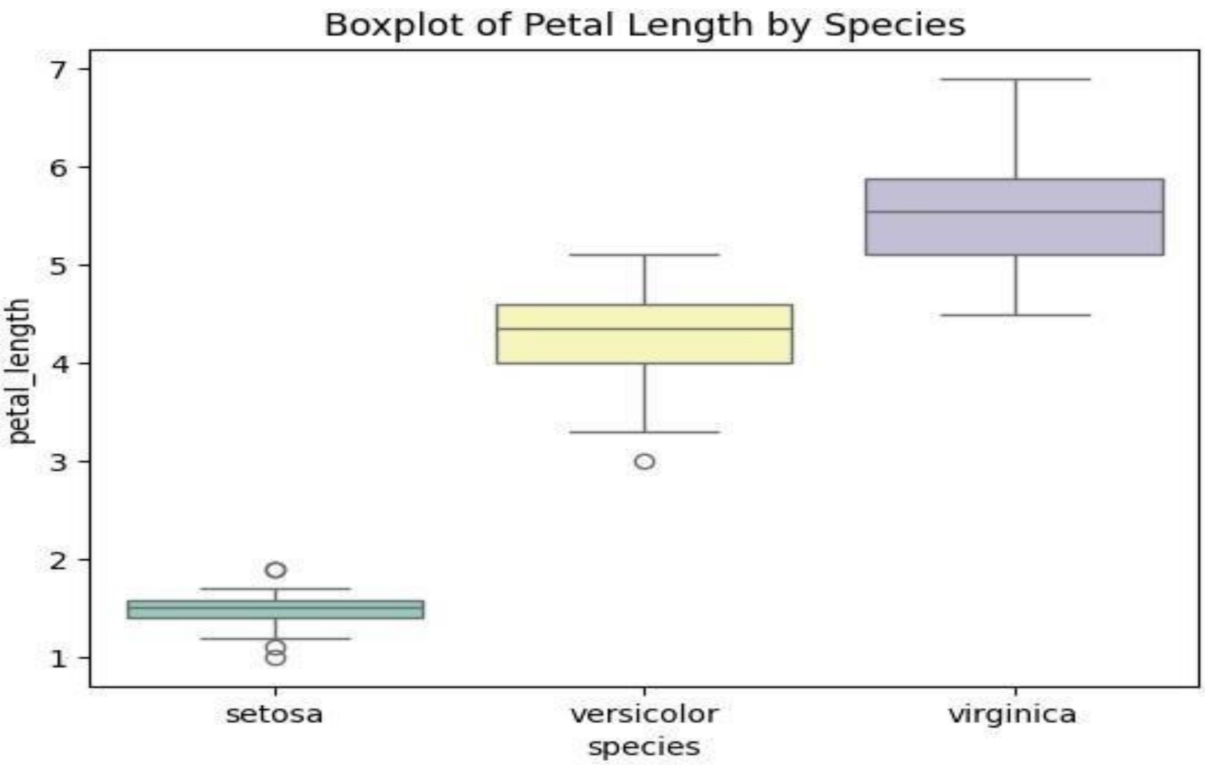



Count of Each Species
count
50
40
30
20
10
0
setosa
versicolor
virginica
species


Sepal Length vs Sepal Width
Sepal Width
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
Sepal Length



Pairplot of Features by Species
sepal_length
sepal_width
petal_length
petal_width
species
setosa
versicolor
virginica



Boxplot of Petal Length by Species
petal_length
setosa
versicolor
virginica
species

1c. Create or Explore datasets to use all pre-processing routines like label encoding, scaling, and binarization.

Code :

1. **Import Necessary Libraries # Import required libraries**

import pandas as pd import numpy as np from sklearn.preprocessing

import LabelEncoder, MinMaxScaler, StandardScaler, Binarizer

2. **Create or Load a Dataset # Create a sample dataset** data

= pd.DataFrame({

'Category': ['A', 'B', 'C', 'A', 'B', 'C'],

 # Categorical variable

'Age': [23, 45, 31, 22, 35, 30],

 # Numerical variable

'Income': [50000, 60000, 70000, 80000, 90000, 100000], #

Numerical variable 'Has_Car':

['Yes', 'No', 'Yes', 'No', 'Yes', 'No']

 # Binary categorical variable })

 # Display the dataset

print("Sample

Dataset:") print(data)

3. **Apply Pre-Processing Routines #**

Label Encoding for 'Category' column

label_encoder = LabelEncoder()

data['Category_Encoded'] =

label_encoder.fit_transform(data['Category']

)

 # Label Encoding for binary column 'Has_Car'

data['Has_Car_Encoded'] =

label_encoder.fit_transform(data['Has_Car']) print("\nAfter Label Encoding:")

print(data)

Min-Max Scaling for 'Income' min_max_scaler

= MinMaxScaler()

data['Income_MinMax'] = min_max_scaler.fit_transform(data[['Income']])

Standard Scaling for 'Age' standard_scaler = StandardScaler()

data['Age_Standardized'] =

standard_scaler.fit_transform(data[['Age']]) print("\nAfter Scaling:")

print(data)

 # Binarization for 'Income' with a threshold of 75,000

binarizer = Binarizer(threshold=75000)

data['Income_Binary'] =

binarizer.fit_transform(data[['Income']]) print("\nAfter Binarization:")

print(data)

4. Save the Processed Dataset # Save the processed dataset

data.to_csv('processed_data.csv', index=False) print("\nProcessed dataset saved as 'processed_data.csv'")

Output :

```
Sample Dataset:
  Category  Age  Income Has_Car
0        A   23   50000     Yes
1        B   45   60000      No
2        C   31   70000     Yes
3        A   22   80000      No
4        B   35   90000     Yes
5        C   30  100000      No
```

```
After Label Encoding:
  Category  Age  Income Has_Car  Category_Encoded  Has_Car_Encoded
0        A   23   50000     Yes                 0                1
1        B   45   60000      No                 1                0
2        C   31   70000     Yes                 2                1
3        A   22   80000      No                 0                0
4        B   35   90000     Yes                 1                1
5        C   30  100000      No                 2                0
```

```
After Scaling:
  Category  Age  Income Has_Car  Category_Encoded  Has_Car_Encoded  \
0        A   23   50000     Yes                 0                1
1        B   45   60000      No                 1                0
2        C   31   70000     Yes                 2                1
3        A   22   80000      No                 0                0
4        B   35   90000     Yes                 1                1
5        C   30  100000      No                 2                0

   Income_MinMax  Age_Standardized
0            0.0         -1.035676
1            0.2          1.812434
2            0.4          0.000000
3            0.6         -1.165136
4            0.8          0.517838
5            1.0         -0.129460
```

```
After Binarization:
  Category  Age  Income Has_Car  Category_Encoded  Has_Car_Encoded  \
0        A   23   50000     Yes                 0                1
1        B   45   60000      No                 1                0
2        C   31   70000     Yes                 2                1
3        A   22   80000      No                 0                0
4        B   35   90000     Yes                 1                1
5        C   30  100000      No                 2                0

   Income_MinMax  Age_Standardized  Income_Binary
0            0.0         -1.035676              0
1            0.2          1.812434              0
2            0.4          0.000000              0
3            0.6         -1.165136              1
4            0.8          0.517838              1
5            1.0         -0.129460              1
```

Practical 2 : Testing Hypothesis

Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a. CSV file and generate the final specific hypothesis. (Create your dataset)

CODE :

```
import pandas as pd

# Step 1: Create the Dataset and Load It data = {'Outlook': ['Sunny', 'Sunny', 'Overcast',
'Rainy', 'Rainy', 'Rainy', 'Overcast', 'Sunny', 'Sunny', 'Rainy'],

'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool', 'Cool', 'Mild', 'Cool', 'Mild'],

'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal', 'Normal', 'High', 'Normal',
'Normal'],

'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak', 'Strong', 'Strong', 'Weak', 'Weak', 'Weak'],

'PlayTennis': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes', 'No', 'Yes', 'Yes']

}


# Load dataset into a pandas DataFrame df
= pd.DataFrame(data)

        # Step 2: Implementing the FIND-S Algorithm def
        find_s_algorithm(data):
          # Get the positive examples (PlayTennis = 'Yes') positive_examples
          = data[data['PlayTennis'] == 'Yes']
          # Initialize hypothesis with the first positive example (most specific) hypothesis
          = positive_examples.iloc[0].drop('PlayTennis')
          # Loop through the rest of the positive examples and generalize
        the hypothesis for index, row in positive_examples.iterrows():
            for feature in hypothesis.index:
              if hypothesis[feature] != row[feature]:
                hypothesis[feature] = '?'
          return hypothesis Step 3:
Apply FIND-S to the dataset
```

hypothesis = find_s_algorithm(df) #

Display the final specific hypothesis

print("The most specific hypothesis is:")

print(hypothesis)

Output :

```
Dataset:
     Sky Temperature Humidity    Wind Water Forecast Condition
0  Sunny        Warm   Normal  Strong  Warm     Same       Yes
1  Sunny        Cold     High  Strong  Warm     Same        No
2  Rainy        Warm     High    Weak  Cool   Change        No
3  Sunny        Warm   Normal  Strong  Warm     Same       Yes
4  Rainy        Cold   Normal    Weak  Cool   Change        No
```

```
Loaded Dataset:
     Sky Temperature Humidity    Wind Water Forecast Condition
0  Sunny        Warm   Normal  Strong  Warm     Same       Yes
1  Sunny        Cold     High  Strong  Warm     Same        No
2  Rainy        Warm     High    Weak  Cool   Change        No
3  Sunny        Warm   Normal  Strong  Warm     Same       Yes
4  Rainy        Cold   Normal    Weak  Cool   Change        No
```

```
Final Specific Hypothesis:
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
```

Practical 3 : Linear Models

3a. Simple Linear Regression

Fit a linear regression model on a dataset. Interpret coefficients, make predictions, and evaluate performance using metrics like R-squared and MSE

Code :

Step 1: Import Libraries # Import required libraries

import pandas as pd import numpy as np import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split from

sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

Step 2: Create a Dataset and Save as CSV

 # Create a sample dataset data

= {

'House_Size': [750, 800, 850, 900, 1000, 1100, 1200, 1300, 1400, 1500],

'Price': [150000, 160000, 165000, 170000, 180000, 190000, 200000, 210000, 220000, 230000]

}

 # Convert the dataset into a DataFrame

df = pd.DataFrame(data)

 # Save to CSV file
df.to_csv('house_prices.csv', index=False)

 # Display the dataset
print("Dataset:")
print(df)

Step 3: Load the Dataset

 # Load the dataset dataset =

pd.read_csv('house_prices.csv')

 # Display the first few rows

print("\nLoaded

Dataset:")

print(dataset.head())

Step 4: Split the Dataset into Training and Test Sets

 # Features and target variable

X = dataset[['House_Size']] **# Feature: House size** y

= dataset['Price'] **# Target: Price**

 # Split data into training and testing sets (80% train, 20% test)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42) print("\nTraining and Testing Data Sizes:") print("Training Data Size:", X_train.shape[0]) print("Testing Data Size:", X_test.shape[0]) **Step 5: Fit a Linear Regression Model**

 # Initialize and fit the linear regression model

model = LinearRegression() model.fit(X_train, y_train) **# Display the coefficients** print("\nModel Coefficients:") print("Slope (m):", model.coef_[0]) print("Intercept (b):", model.intercept_)

Step 6: Make Predictions #

Predict on the test set

y_pred = model.predict(X_test) # **Display predictions** print("\nPredictions on Test Data:") print("Actual Prices:", y_test.values) print("Predicted Prices:", y_pred)

Step 7: Evaluate the Model # Calculate evaluation metrics mse =

mean_squared_error(y_test, y_pred) r2 = r2_score(y_test, y_pred) #

Display metrics

print("\nModel Performance Metrics:")
print("Mean Squared Error (MSE):", mse)
print("R-squared (R²):", r2)

Step 8: Visualize the Results # Scatter plot of the training data plt.scatter(X_train, y_train, color='blue', label='Training Data')

 # Plot the regression line plt.plot(X_train, model.predict(X_train),

color='red', label='Regression Line')

 # Scatter plot of the test data

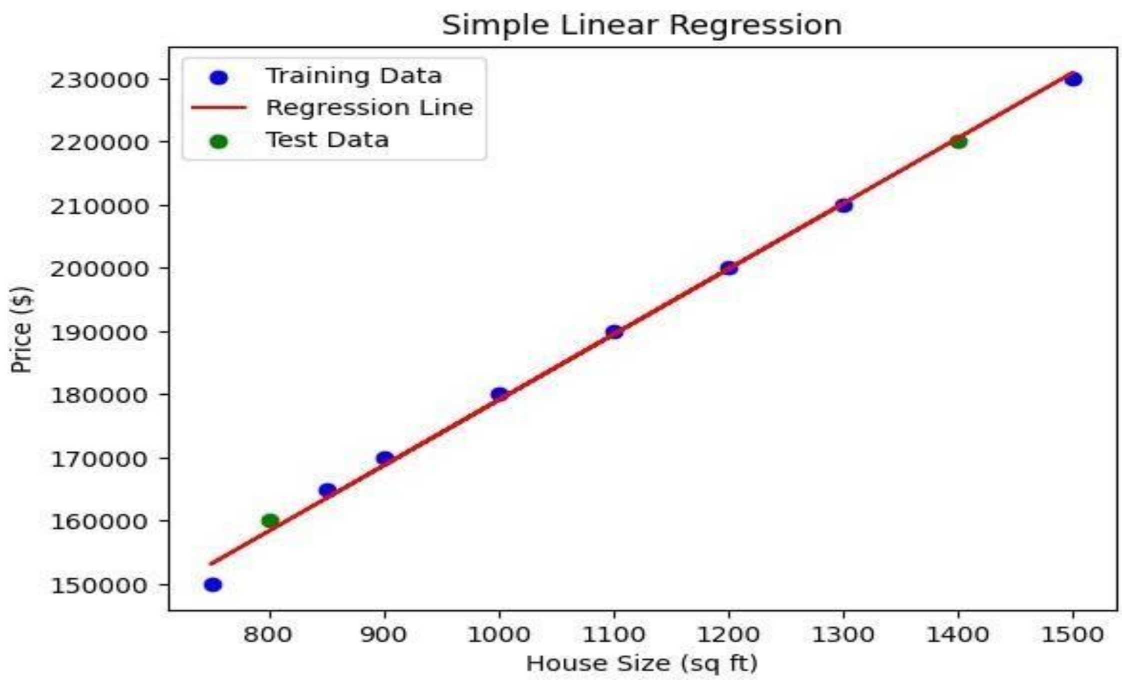
plt.scatter(X_test, y_test, color='green', label='Test Data')

plt.title("Simple Linear Regression") plt.xlabel("House

Size (sq ft)") plt.ylabel("Price ($)") plt.legend()

plt.show()

Output :



3b. Multiple Linear Regression :

Extend linear regression to multiple feature. Handle feature selection and potential multicollinearity

Code :

import pandas as pd import

numpy as np import

matplotlib.pyplot as plt import

seaborn as sns

from sklearn.model_selection import train_test_split from

sklearn.linear_model import LinearRegression from

sklearn.metrics import mean_squared_error, r2_score

from statsmodels.stats.outliers_influence import variance_inflation_factor from
sklearn.preprocessing import LabelEncoder

 # Import LabelEncoder

from sklearn.impute import SimpleImputer

 # Load dataset from google.colab import files uploaded = files.upload() **#**

Upload your CSV file

 # Read the CSV file data

=pd.read_csv(list(uploaded.keys())[0])

 # Display the first few rows print(data.head())

 # Check for null values and basic statistics
print(data.info())
print(data.describe())

 # Define a function to calculate VIF def
calculate_vif(df):

 # Select only numeric features for VIF calculation numeric_df
= df.select_dtypes(include=np.number)

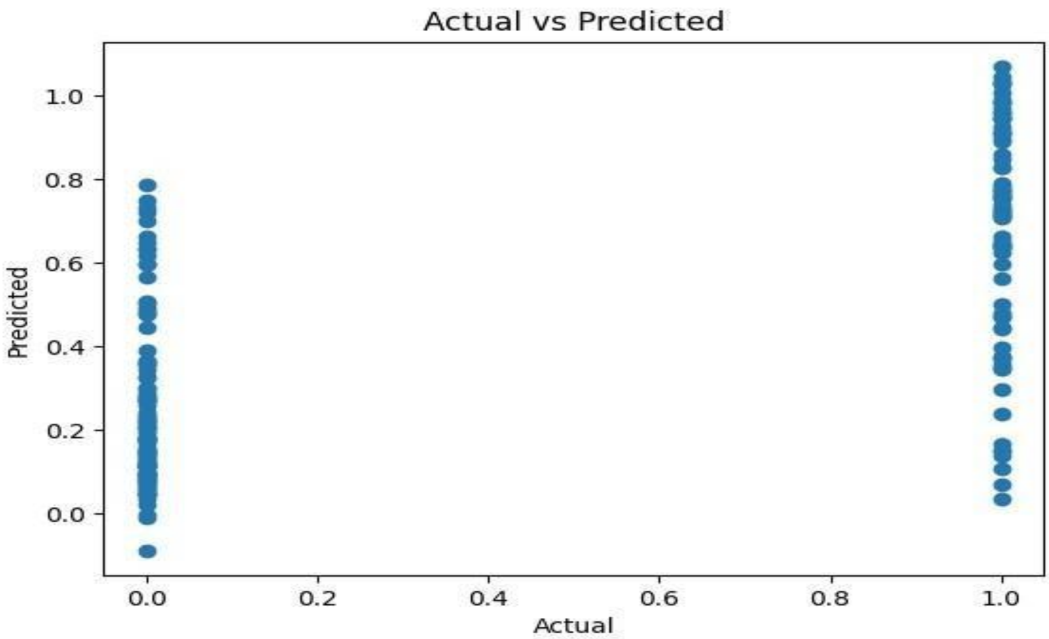
 # Drop rows with infinite or missing values

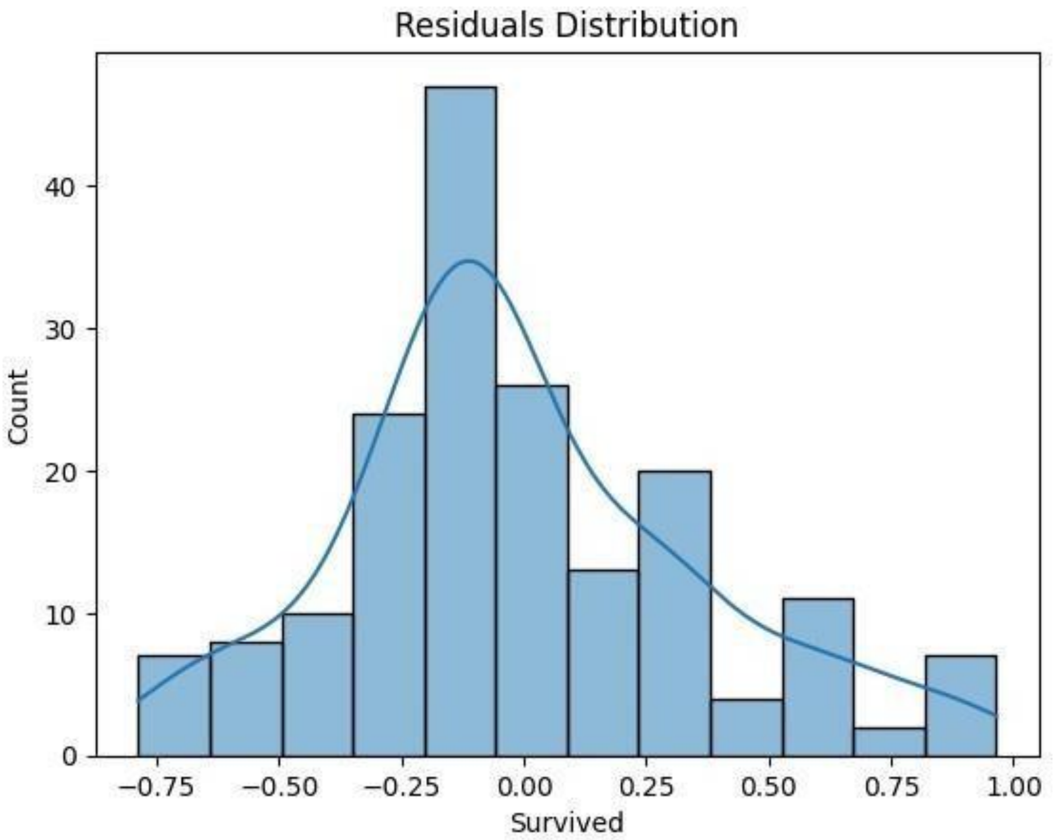
numeric_df = numeric_df.replace([np.inf, -np.inf], np.nan).dropna() vif_data
= pd.DataFrame()
vif_data["feature"] = numeric_df.columns
vif_data["VIF"] = [variance_inflation_factor(numeric_df.values,
i) for i in range(numeric_df.shape[1])] return vif_data

```
# Selecting features and target variable
X = data.drop("Survived", axis=1)
# Changed 'y' to 'Survived' y = data["Survived"]

# Handle categorical features (e.g., using Label Encoding)
for col in X.select_dtypes(include=['object']).columns: le =
LabelEncoder()
   X[col] = le.fit_transform(X[col])

# Impute missing values using the mean (you can choose other strategies) imputer
= SimpleImputer(strategy='mean')
# Create an imputer instance
X = pd.DataFrame(imputer.fit_transform(X), columns=X.columns)
# Impute and update X

# Calculate VIF for initial features print("VIF
before handling multicollinearity:")
print(calculate_vif(X)) # Call the modified function

# Drop features based on VIF analysis (example: drop 'X1' if VIF is high)
# Check if the column exists before dropping if
'X1' in X.columns:
   X = X.drop("X1", axis=1) # Replace 'X1' with the actual high VIF feature name else:
   print("Column 'X1' not found in the DataFrame.")

# Recalculate VIF
print("VIF after handling multicollinearity:")
print(calculate_vif(X))
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize and fit the model
model = LinearRegression()
model.fit(X_train, y_train)

# Get coefficients and intercept
print("Coefficients:", model.coef_)
print("Intercept:",
model.intercept_)

# Predictions
y_pred = model.predict(X_test)

# Evaluation metrics
rmse = np.sqrt(mean_squared_error(y_test, y_pred)) r2 = r2_score(y_test, y_pred)
print(f"RMSE: {rmse}") print(f"R^2: {r2}")
from sklearn.feature_selection import RFE
```

```
# Recursive Feature Elimination
rfe = RFE(estimator=LinearRegression(), n_features_to_select=5)

# Adjust features
rfe.fit(X_train, y_train)

# Selected features
print("Selected Features:", X.columns[rfe.support_])

# Scatter plot of actual vs predicted values
plt.scatter(y_test,                     y_pred)
plt.xlabel("Actual")     plt.ylabel("Predicted")
plt.title("Actual vs Predicted") plt.show()

# Residuals
residuals = y_test - y_pred sns.histplot(residuals, kde=True)
plt.title("Residuals Distribution") plt.show()
```

Output :




Residuals Distribution
Count
0
10
20
30
40
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Survived

3c. Regualarized Linear Models :

Implement Regression variants like LASSO and Ridge on any generated dataset

Code :

1. **Set Up the Environment**

import numpy as np import

pandas as pd import

matplotlib.pyplot as plt import

seaborn as sns

from sklearn.model_selection import train_test_split from

sklearn.linear_model import Ridge, Lasso, ElasticNet from

sklearn.metrics import mean_squared_error, r2_score from

sklearn.datasets import make_regression # **Set random**

seed for reproducibility

np.random.seed(42)

2. **Generate a Synthetic Dataset**

 # Generate synthetic data

X, y = make_regression(n_samples=1000,

 # Number of samples

n_features=10, #

Number of features

noise=15,

 # Add some noise

random_state=42

)

 # Convert to DataFrame for exploration

data = pd.DataFrame(X, columns=[f"X{i}"

for i in range(1, 11)]) data["y"] = y

 # Display the first few rows

print(data.head())

3. **Split the Dataset**

 # Split data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(data.drop("y", axis=1),

 # Features data["y"],

 # Target variable test_size=0.2,

 # 20% for testing random_state=42

)

4. **Train and Evaluate Ridge Regression**

 # Initialize Ridge Regression with a regularization parameter (alpha)

ridge = Ridge(alpha=1.0) # **Train the model** ridge.fit(X_train, y_train) #

Predictions

ridge_pred = ridge.predict(X_test) #

Evaluate Ridge Regression

ridge_rmse = np.sqrt(mean_squared_error(y_test, ridge_pred))

ridge_r2 = r2_score(y_test, ridge_pred) print(f"Ridge RMSE:

{ridge_rmse}") print(f"Ridge R^2: {ridge_r2}")

5. **Train and Evaluate Lasso Regression # Initialize Lasso Regression** lasso = Lasso(alpha=0.1)

 # Train the model lasso.fit(X_train,

y_train) # **Predictions**

lasso_pred = lasso.predict(X_test)

Evaluate Lasso Regression lasso_rmse = np.sqrt(mean_squared_error(y_test, lasso_pred)) lasso_r2 = r2_score(y_test, lasso_pred) print(f"Lasso RMSE: {lasso_rmse}") print(f"Lasso R^2: {lasso_r2}") # **Features shrunk to**

zero print("Lasso Coefficients:", lasso.coef_)

6. **Train and Evaluate ElasticNet Regression # Initialize ElasticNet**

elastic_net = ElasticNet(alpha=0.1, l1_ratio=0.5) # l1_ratio balances L1 and L2 penalties

 # Train the model elastic_net.fit(X_train,

y_train)

 # Predictions

elastic_net_pred = elastic_net.predict(X_test)

 # Evaluate

ElasticNet Regression elastic_net_rmse = np.sqrt(mean_squared_error(y_test, elastic_net_pred)) elastic_net_r2 = r2_score(y_test, elastic_net_pred) print(f"ElasticNet RMSE: {elastic_net_rmse}") print(f"ElasticNet R^2: {elastic_net_r2}")

7. **Compare Results**

 # Collect metrics metrics
= pd.DataFrame({

"Model": ["Ridge", "Lasso", "ElasticNet"],

"RMSE": [ridge_rmse, lasso_rmse, elastic_net_rmse],

"R^2": [ridge_r2, lasso_r2, elastic_net_r2]
})

print(metrics)

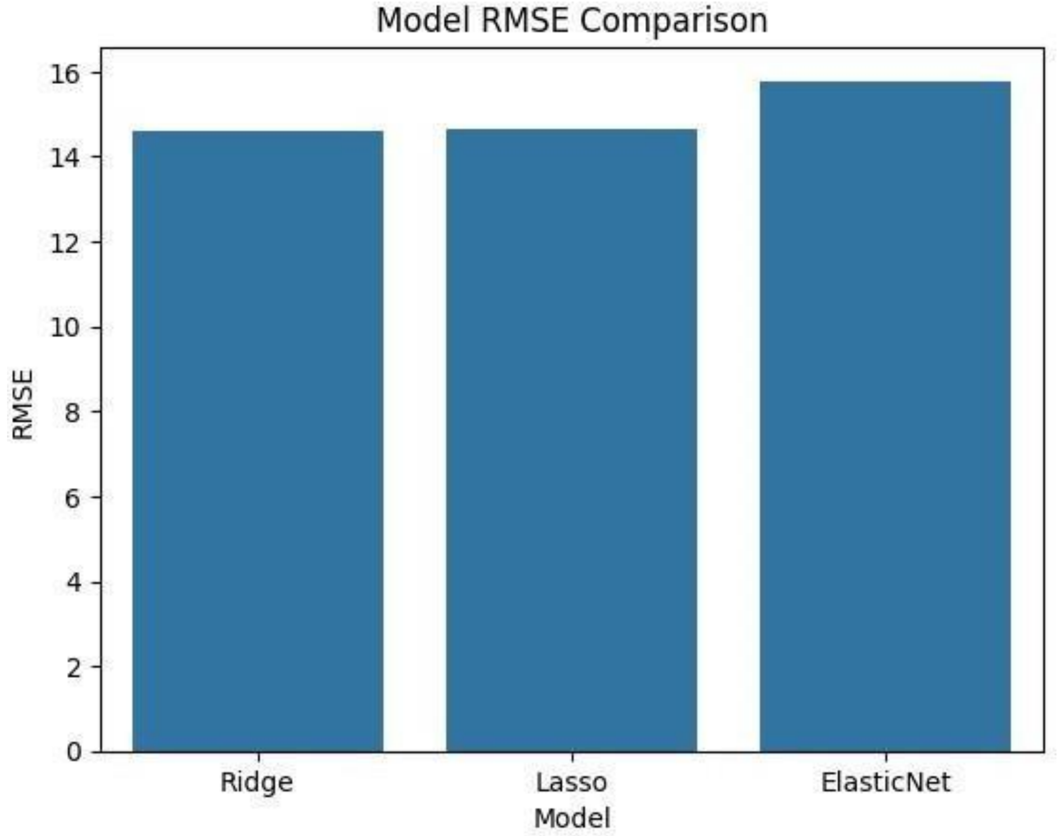
 # Plot RMSE comparison

sns.barplot(data=metrics, x="Model",

y="RMSE") plt.title("Model RMSE Comparison")

plt.show()

Output :



Model RMSE Comparison
16
14
12
10
8
6
4
2
0
RMSE
Ridge
Lasso
ElasticNet
Model

Practical 4 : Discriminative Models

4a. Logistic Regression :

Perform binary classification using logistic regression. Calculate accuracy, precision, recall, and understand the ROC curve."

Code :

Step 1: Import Required Libraries # Import necessary libraries import numpy as np import pandas as pd from sklearn.model_selection import train_test_split from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, precision_score, recall_score, roc_curve, auc

import matplotlib.pyplot as plt

Step 2: Prepare the Dataset from sklearn.datasets import make_classification

 # Create a synthetic dataset

X, y = make_classification(n_samples=1000, n_features=10, n_classes=2, random_state=42)

 # Split data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

Step 3: Train the Logistic Regression Model

 # Initialize the logistic regression model logreg = LogisticRegression()

 # Train the model on the training data logreg.fit(X_train, y_train)

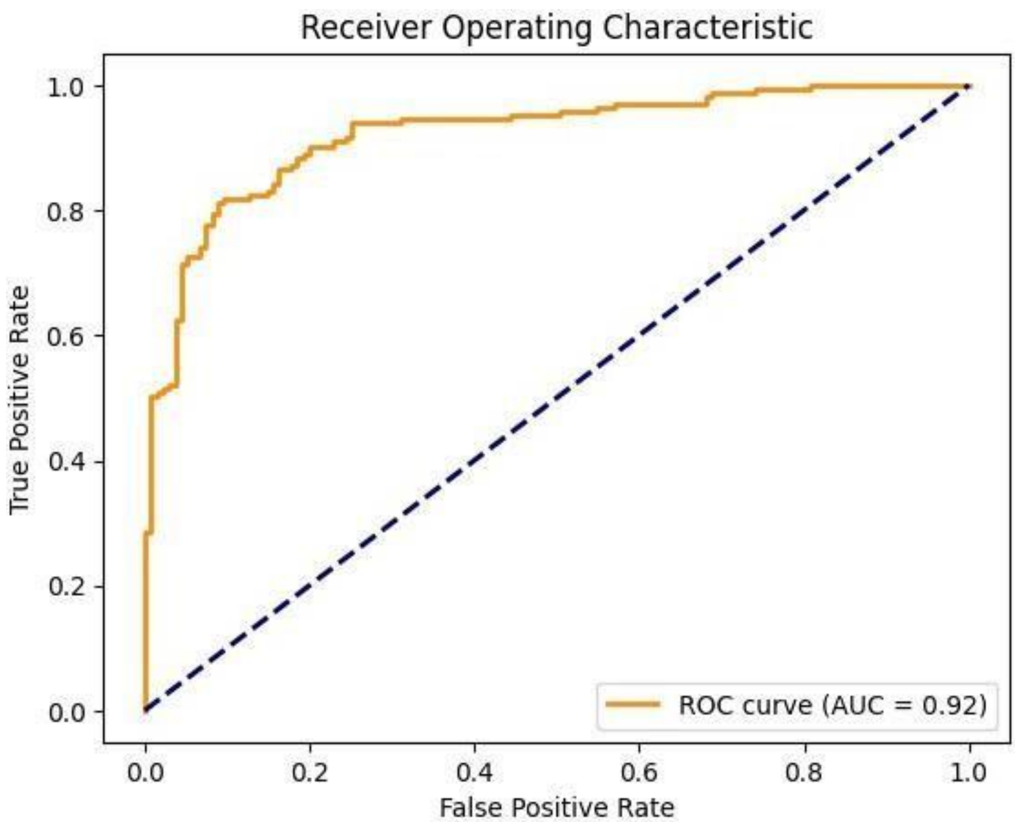
Step 4: Make Predictions

 # Predict labels for the test set y_pred = logreg.predict(X_test)

 # Predict probabilities for the ROC curve
y_prob = logreg.predict_proba(X_test)[:, 1]

Step 5: Evaluate the Model # Calculate metrics accuracy = accuracy_score(y_test, y_pred) precision = precision_score(y_test, y_pred) recall = recall_score(y_test, y_pred) print(f"Accuracy: {accuracy:.2f}")

Output :



4b .Implement and demonstrate k-nearest Neighbor algorithm. Read the training data from a .CSV file and build the model to classify a test sample. Print both correct and wrong predictions.

Code :

Step 1: Import Required Libraries # Import necessary libraries

import pandas as pd import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

from google.colab import files

Step 2: Create or Upload the CSV File

 # Check if the user wants to create a dataset or upload one

print("Do you have a CSV file to upload? (yes/no)") response
= input().lower() if response == "yes":

uploaded = files.upload()
filename = list(uploaded.keys())[0]
else:

 # Create a synthetic dataset from

sklearn.datasets import make_classification

 # Generate synthetic data

X,y=make_classification(n_samples=200,n_features=5, n_classes=2, random_state=42)

 # Combine features and target into a single DataFrame

data = pd.DataFrame(X, columns=[f"Feature_{i}" for i in

range(X.shape[1])]) data['Target'] = y

 # Save the dataset to a CSV file filename =

"synthetic_data.csv" data.to_csv(filename,

index=False) print(f"Synthetic dataset saved as

{filename}.") **Step 3: Load the CSV File into**

a DataFrame

# Load the dataset into a DataFrame
data = pd.read_csv(filename)
 # Display the first few rows of the dataset
print("Loaded Dataset:")
print(data.head())

Step 4: Preprocess the Data

 # Separate features (X) and labels (y)

X = data.iloc[:, :-1].values # All columns except the last one
y = data.iloc[:, -1].values # Last column as the target

 # Split the dataset into training and testing sets (80% train, 20% test)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 5: Train the k-NN Model #

Initialize the k-NN model with k=3 knn

= KNeighborsClassifier(n_neighbors=3) #

Train the model on the training data

knn.fit(X_train, y_train)

Step 6: Predict Test Samples #

Predict the labels for the test set

y_pred = knn.predict(X_test)

Step 7: Evaluate and Print Predictions #
Calculate and display the accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"\nModel Accuracy:
{accuracy:.2f}\n")

 # Display correct and incorrect predictions
print("Correct Predictions:") for
i in range(len(y_test)):
if y_pred[i] == y_test[i]: print(f"Sample {i}:
Predicted={y_pred[i]}, Actual={y_test[i]}") print("\nIncorrect
Predictions:") for i in range(len(y_test)):
if y_pred[i] != y_test[i]:
print(f"Sample {i}: Predicted={y_pred[i]}, Actual={y_test[i]}")

Output :

```
Model Accuracy: 0.88

Correct Predictions:
Sample 0: Predicted=0, Actual=0
Sample 1: Predicted=1, Actual=1
Sample 2: Predicted=1, Actual=1
Sample 3: Predicted=0, Actual=0
Sample 4: Predicted=1, Actual=1
Sample 5: Predicted=1, Actual=1
Sample 6: Predicted=0, Actual=0
Sample 7: Predicted=0, Actual=0
Sample 9: Predicted=1, Actual=1
Sample 10: Predicted=1, Actual=1
Sample 11: Predicted=1, Actual=1
Sample 12: Predicted=0, Actual=0
Sample 13: Predicted=0, Actual=0
Sample 14: Predicted=0, Actual=0
Sample 15: Predicted=0, Actual=0
Sample 16: Predicted=0, Actual=0
Sample 17: Predicted=1, Actual=1
Sample 18: Predicted=1, Actual=1
Sample 19: Predicted=0, Actual=0
Sample 20: Predicted=0, Actual=0
Sample 22: Predicted=1, Actual=1
Sample 23: Predicted=1, Actual=1
Sample 24: Predicted=1, Actual=1
Sample 25: Predicted=1, Actual=1
Sample 26: Predicted=1, Actual=1
Sample 27: Predicted=0, Actual=0
Sample 28: Predicted=0, Actual=0
Sample 30: Predicted=1, Actual=1
Sample 31: Predicted=1, Actual=1
Sample 32: Predicted=1, Actual=1
Sample 34: Predicted=0, Actual=0
Sample 35: Predicted=1, Actual=1
Sample 36: Predicted=1, Actual=1
Sample 38: Predicted=1, Actual=1
Sample 39: Predicted=1, Actual=1
```

```
Incorrect Predictions:
Sample 8: Predicted=1, Actual=0
Sample 21: Predicted=1, Actual=0
Sample 29: Predicted=0, Actual=1
Sample 33: Predicted=1, Actual=0
Sample 37: Predicted=1, Actual=0
```

4c. Build a decision tree classifier or regressor. Control hyperparameters like tree depth to avoid overfitting. Visualize the tree.

Code :

Step 1: Import Required Libraries

 # Import necessary libraries

import pandas as pd import numpy

as np

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier, DecisionTreeRegressor, plot_tree

from sklearn.metrics import accuracy_score, mean_squared_error import

matplotlib.pyplot as plt

from google.colab import files

Step 2: Create or Upload the CSV File

 # Check if the user wants to upload a file or generate one

print("Do you have a CSV file to upload? (yes/no)") response

= input().lower() if response == "yes":

 # Upload the CSV file uploaded =

files.upload() filename =

list(uploaded.keys())[0]

else:

 # Generate synthetic data (classification or regression)

from sklearn.datasets import make_classification, make_regression

print("Choose a task: (1) Classification (2) Regression")

task = int(input()) if task == 1:

 # Generate synthetic classification data

X, y = make_classification(n_samples=200, n_features=5, random_state=42) task_type

= "classification"

else:

 # Generate synthetic regression data

X, y = make_regression(n_samples=200, n_features=5, random_state=42) task_type

= "regression"

 # Combine features and target into a single DataFrame

data = pd.DataFrame(X, columns=[f"Feature_{i}" for i in

range(X.shape[1])]) data['Target'] = y

 # Save the dataset to a CSV file

filename = "synthetic_data.csv"

data.to_csv(filename, index=False)

print(f"Synthetic {task_type} dataset saved as {filename}.")

Step 3: Load the Dataset #

Load the dataset data

=pd.read_csv(filename)

 # Display the first few rows of the dataset

print("Dataset Preview:") print(data.head())

Step 4: Preprocess the Data

 # Separate features and target

X = data.iloc[:, :-1].values # All columns except the last one
y = data.iloc[:, -1].values # Last column as the target

 # Split data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 5: Build the Decision Tree

 # Define the tree depth to avoid overfitting max_depth

= 3

 # Initialize the model

if task_type =="classification": model =
DecisionTreeClassifier(max_depth=max_depth, random_state=42) else:

model = DecisionTreeRegressor(max_depth=max_depth, random_state=42)

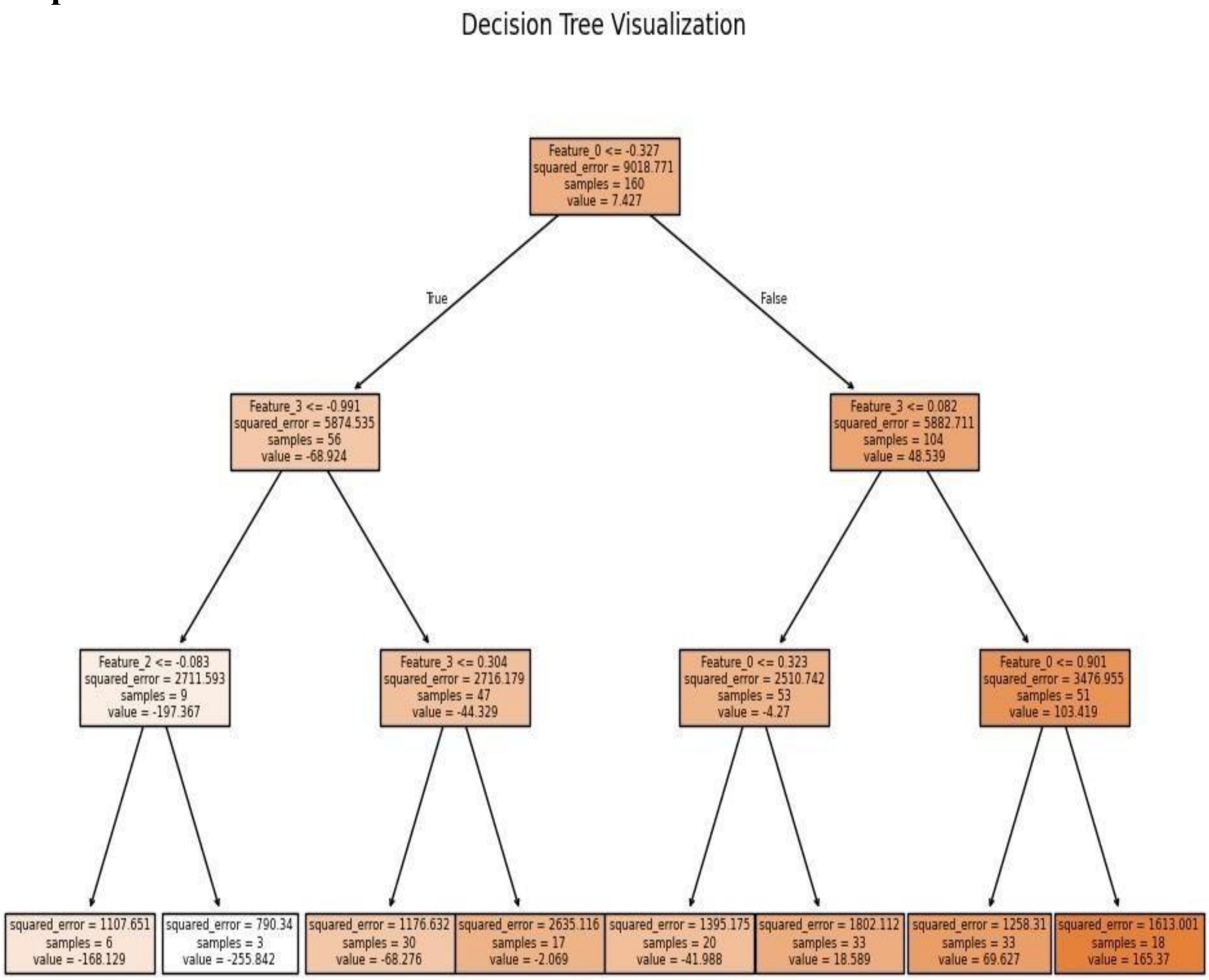
 # Train the model
model.fit(X_train, y_train)

Step 6: Make Predictions #
Predict on the test set y_pred =
model.predict(X_test) **#**
Evaluate the model if task_type
== "classification":

accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.2f}")
else:
mse = mean_squared_error(y_test, y_pred)
print(f"Mean Squared Error: {mse:.2f}")

Step 7: Visualize the Tree # Visualize the decision tree plt.figure(figsize=(12, 8)) plot_tree(model, feature_names=data.columns[:-1], class_names=str(np.unique(y)) if task_type == "classification" else None, filled=True) plt.title("Decision Tree Visualization") plt.show()

Output :



4d. Implement a Support Vector Machine for any relevant dataset.

Code:

Step 1: Import Required Libraries # Import necessary libraries import pandas as pd import numpy as np from sklearn.model_selection import train_test_split from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, classification_report from google.colab import files

Step 2: Create or Upload a Dataset

 # Check if the user wants to upload a file or generate one

print("Do you have a CSV file to upload? (yes/no)") response = input().lower() if response == "yes": **# Upload the CSV file** uploaded = files.upload() filename = list(uploaded.keys())[0] else:

 # Generate synthetic classification data from sklearn.datasets import make_classification

X, y = make_classification(n_samples=200, n_features=5, n_classes=2, random_state=42)

 # Combine features and target into a DataFrame

data = pd.DataFrame(X, columns=[f"Feature_{i}" for i in range(X.shape[1])]) data['Target'] = y

#Save the synthetic dataset to a CSV file

filename="synthetic_data.csv"

data.to_csv(filename,index=False)

print(f"Synthetic dataset saved as {filename}.")

Step 3: Load the Dataset

 # Load the dataset into a DataFrame data = pd.read_csv(filename)

 # Display the first few rows of the dataset
print("Dataset Preview:") print(data.head())

Step 4: Preprocess the Data

 # Separate features (X) and target (y)

X = data.iloc[:, :-1].values # All columns except the last one
y = data.iloc[:, -1].values # Last column as the target

 # Split the dataset into training (80%) and testing (20%) sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 5: Train the Support Vector Machine

 # Initialize the SVM model (use RBF kernel as default) svm_model = SVC(kernel='rbf', C=1.0, gamma='scale', random_state=42)

 # Train the SVM model on the training data

svm_model.fit(X_train, y_train)

Step 6: Make Predictions

 # Predict the labels for the test set

y_pred = svm_model.predict(X_test)

Step 7: Evaluate the Model # Calculate and print the accuracy accuracy = accuracy_score(y_test, y_pred)

print(f"Model Accuracy: {accuracy:.2f}")

 # Print a detailed classification report

print("\nClassification Report:")

print(classification_report(y_test, y_pred))

Step 8: Visualize the Decision Boundary (Optional for 2D Data) import matplotlib.pyplot as plt

 # Generate 2D synthetic data from sklearn.datasets import make_blobs

X, y = make_blobs(n_samples=100, centers=2, random_state=42, cluster_std=1.5)

 # Fit the SVM on this data svm_model.fit(X, y)

#Plot the decision boundary plt.figure(figsize=(8, 6))

plt.scatter(X[:, 0], X[:, 1], c=y, cmap='coolwarm', edgecolor='k')

 # Create a grid to evaluate the model xx, yy = np.meshgrid (np.linspace(X[:, 0].min(), X[:, 0].max(), 100), np.linspace(X[:, 1].min(), X[:, 1].max(), 100))

Z = svm_model.decision_function(np.c_[xx.ravel(), yy.ravel()])

Z = Z.reshape(xx.shape)

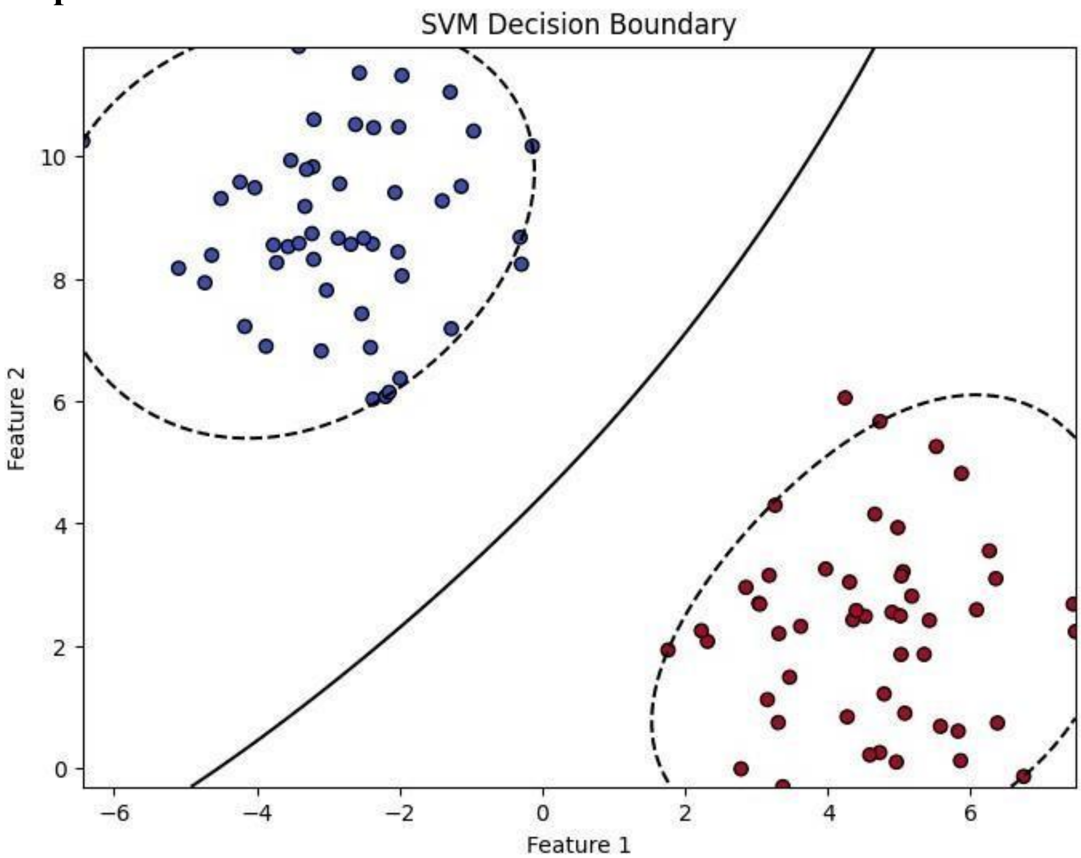
 # Plot the decision boundary and margins

plt.contour(xx, yy, Z, levels=[-1, 0, 1], linestyles=['--', '-', '--'], colors='k')

plt.title("SVM Decision Boundary") plt.xlabel("Feature 1")

plt.ylabel("Feature 2") plt.show()

Output :



4e. Train a random forest ensemble. Experiment with the number of trees and feature sampling. Compare performance to a single decision tree.

Code :

Step 1: Import Required Libraries # Import necessarylibraries

import pandas as pd import numpy as np

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier from

sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, classification_report from
google.colab import files

Step 2: Create or Upload a Dataset

 # Check if the user wants to upload a file or generate one
print("Do you have a CSV file to upload? (yes/no)")
response = input().lower() if response == "yes": **# Upload the CSV file** uploaded = files.upload() filename = list(uploaded.keys())[0] else:
 # Generate synthetic classification data from

sklearn.datasets import make_classification

X, y = make_classification(n_samples=300, n_features=10, n_classes=2, random_state=42)

 # Combine features and target into a DataFrame

data = pd.DataFrame(X, columns=[f"Feature_{i}"

for i in range(X.shape[1])]) data['Target'] = y

 # Save the synthetic dataset to a CSV file
filename = "synthetic_data.csv"
data.to_csv(filename, index=False)
print(f"Synthetic dataset saved as {filename}.")

Step 3: Load the Dataset

 # Load the dataset

data =pd.read_csv(filename)

 # Display the first few rows of the dataset print("Dataset
Preview:")
print(data.head())

Step 4: Preprocess the Data

 # Separate features (X) and target (y)

X = data.iloc[:, :-1].values # All columns except the last one
y = data.iloc[:, -1].values # Last column as the target

 # Split the dataset into training (80%) and testing (20%) sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 5: Train a Single Decision Tree Classifier # Initialize and train the Decision Tree model
decision_tree =DecisionTreeClassifier(random_state=42)
decision_tree.fit(X_train, y_train) #
Predict and evaluate
y_pred_tree = decision_tree.predict(X_test)
accuracy_tree = accuracy_score(y_test, y_pred_tree)
print(f"Decision Tree Accuracy: {accuracy_tree:.2f}")

Step 6: Train a Random Forest Classifier

 # Initialize the Random Forest model with hyperparameter tuning
random_forest = RandomForestClassifier(n_estimators=100, max_features='sqrt', random_state=42) **# Train the model**
random_forest.fit(X_train, y_train)

 # Predict and evaluate

y_pred_rf = random_forest.predict(X_test) accuracy_rf

= accuracy_score(y_test, y_pred_rf)

print(f"Random Forest Accuracy (100 trees, sqrt features): {accuracy_rf:.2f}")

Step 7: Experiment with Random Forest Hyperparameters # Experiment with fewer trees and different feature sampling
rf_experiment = RandomForestClassifier(n_estimators=50, max_features=3, random_state=42)

rf_experiment.fit(X_train, y_train)
 # Predict and evaluate

y_pred_rf_exp = rf_experiment.predict(X_test) accuracy_rf_exp

= accuracy_score(y_test, y_pred_rf_exp)

print(f"Random Forest Accuracy (50 trees, max_features=3): {accuracy_rf_exp:.2f}")

Step 8: Compare the Models print("\nModel

Comparison:")

print(f"Decision Tree Accuracy: {accuracy_tree:.2f}") print(f"Random

Forest Accuracy (100 trees): {accuracy_rf:.2f}")

print(f"Random Forest Accuracy (50 trees, max_features=3): {accuracy_rf_exp:.2f}")

Step 9: Visualize Feature Importance (Optional) import

matplotlib.pyplot as plt

 # Extract feature importance from the Random Forest model

feature_importances = random_forest.feature_importances_

 # Plot the feature importance plt.figure(figsize=(10,

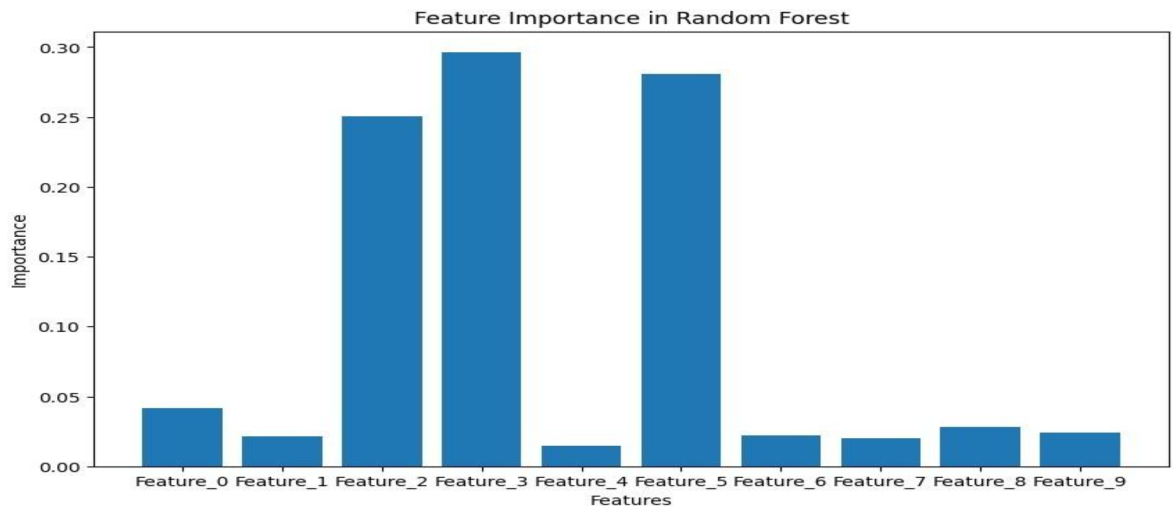
6))

plt.bar(range(len(feature_importances)), feature_importances, tick_label=data.columns[:-1])

plt.title("Feature Importance in Random Forest") plt.xlabel("Features")

plt.ylabel("Importance") plt.show()

Output :



4f. Implement a gradient boosting machine (e.g., XGBoost). Tune hyperparameters and explore feature importance.

Code :

Step 1: Import Required Libraries

 # Import necessary libraries

import pandas as pd import numpy

as np

from sklearn.model_selection import train_test_split,GridSearchCV from sklearn.metrics import accuracy_score, classification_report from xgboost import XGBClassifier, plot_importance import matplotlib.pyplot as plt from google.colab import files

Step 2: Create or Upload a Dataset

 # Check if the user wants to upload a file or generate one print("Do you have a CSV file to upload? (yes/no)")

response = input().lower() if response == "yes": # Upload the CSV file uploaded=files.upload() filename = list(uploaded.keys())[0] else:

 # Generate synthetic classification data from sklearn.datasets import make_classification

X, y = make_classification(n_samples=300, n_features=10, n_classes=2, random_state=42)

 # Combine features and target into a DataFrame

data = pd.DataFrame(X, columns=[f"Feature_{i}"

for i in range(X.shape[1])])data['Target'] = y #

Save the synthetic dataset to a CSV file

filename="synthetic_data.csv"

data.to_csv(filename,index=False)

print(f"Synthetic dataset saved as {filename}.")

Step 3: Load the Dataset # Load the dataset data = pd.read_csv(filename) # **Display the first few rows of the dataset** print("Dataset Preview:")
print(data.head())

Step 4: Preprocess the Data

 # Separate features (X) and target (y)

X = data.iloc[:, :-1].values # All columns except the last one
y = data.iloc[:, -1].values # Last column as the target

 # Split the dataset into training (80%) and testing (20%) sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 5: Train a Basic XGBoost Model

 # Initialize and train the XGBoost model with default parameters

xgb = XGBClassifier(random_state=42) xgb.fit(X_train, y_train)

 # Predict and evaluate the model y_pred

= xgb.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print(f"XGBoost Accuracy (Default Parameters): {accuracy:.2f}") **Step 6: Tune Hyperparameters with GridSearchCV**

 # Define a grid of hyperparameters
param_grid = { 'n_estimators': [50, 100, 150],'learning_rate': [0.01, 0.1, 0.2], 'max_depth': [3, 5, 7] }
 # Initialize GridSearchCV

grid_search = GridSearchCV(estimator=XGBClassifier(random_state=42),param_grid=param_grid, scoring='accuracy', cv=3,verbose=1) **# Fit the model with grid search** grid_search.fit(X_train, y_train)
 # Best parameters from GridSearch

print(f"Best Parameters: {grid_search.best_params_}") **# Train the final model with the best parameters** best_xgb = grid_search.best_estimator_
 # Predict and evaluate y_pred_best =

best_xgb.predict(X_test) accuracy_best =

accuracy_score(y_test, y_pred_best)

print(f"XGBoost Accuracy (Tuned Parameters): {accuracy_best:.2f}")

Step 7: Explore Feature Importance

 # Plot feature importance for the tuned model plt.figure(figsize=(10, 6))

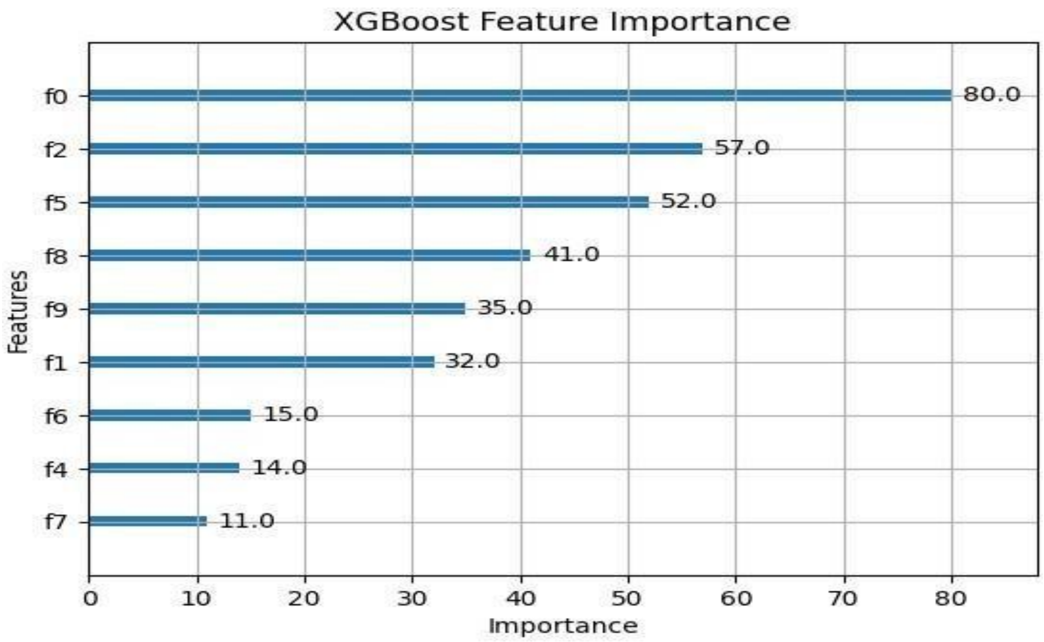
plot_importance(best_xgb,importance_type='weight', xlabel="Importance",

ylabel="Features") plt.title("XGBoost Feature Importance") plt.show()

Step 8: Evaluate the Model

 # Print a detailed classification report
print("Classification Report:")
print(classification_report(y_test, y_pred_best))

Output :



<u>Practical 5</u>

5a. Implement and demonstrate the working of a Naive Bayesian classifier using a sample data set. Build the model to classify a test sample.

Step 1: Import Required Libraries # Import necessary libraries import pandas as pd import numpy as np from sklearn.model_selection import train_test_split from sklearn.metrics import accuracy_score,classification_report from sklearn.naive_bayes import GaussianNB

from google.colab import files

Step 2: Create or Upload a Dataset

 # Ask if the user wants to upload a file or generate one

print("Do you have a CSV file to upload? (yes/no)") response = input().lower() if response == "yes": **# Upload the CSV file** uploaded =files.upload() filename = list(uploaded.keys())[0]

else:

 # Generate synthetic classification data from sklearn.datasets import make_classification

X, y = make_classification(n_samples=300, n_features=8, n_classes=2, random_state=42)

 # Combine features and target into a DataFrame

data = pd.DataFrame(X, columns=[f"Feature_{i}" for i in range(X.shape[1])])
data['Target'] = y

 # Save the synthetic dataset to a CSV file

filename="synthetic_naive_bayes_data.csv"

data.to_csv(filename,index=False)

print(f"Synthetic dataset saved as {filename}.") **Step 3: Load the Dataset # Load the dataset** data =pd.read_csv(filename)

 # Display the first few rows of the dataset

print("Dataset Preview:") print(data.head())

Step 4: Preprocess the Data

 # Separate features (X) and target (y)

X = data.iloc[:, :-1].values # All columns except the last one
y = data.iloc[:, -1].values # Last column as the target

 # Split the dataset into training (80%) and testing (20%) sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 5: Train a Naive Bayes Classifier

 # Initialize the Gaussian Naive Bayes classifier naive_bayes = GaussianNB() **# Train the model**

naive_bayes.fit(X_train, y_train)

Step 6: Make Predictions and Evaluate

 # Predict on the test set

y_pred =naive_bayes.predict(X_test)
 # Evaluate the model

accuracy = accuracy_score(y_test, y_pred)
print(f"Naive Bayes Accuracy: {accuracy:.2f}")

 # Detailed classification report

print("Classification Report:")

print(classification_report(y_test,

y_pred))

Step 7: Test the Model with a Custom Sample

 # Define a sample test input (replace with meaningful values based on your dataset) test_sample = [X_test[0]]

 # Taking the first test sample for demonstration

 # Predict the class for the test sample

predicted_class =

naive_bayes.predict(test_sample) print(f"Test

Sample: {test_sample}") print(f"Predicted Class:

{predicted_class[0]}")

Output :

```
Dataset Preview:
   Feature_0  Feature_1  Feature_2  Feature_3  Feature_4  Feature_5  \
0  -1.274158   1.317988  -2.423879   0.906946  -1.583903  -0.331811
1   1.607963  -1.649959   0.299293  -0.891720   1.301741   1.508502
2  -0.154167   0.161033   2.210523   0.139400  -0.557492   0.087713
3  -0.920991   0.949136  -1.613561   0.588410   1.471170  -0.529287
4   1.013304  -1.038578  -0.305225  -0.539334  -0.609512   1.048078

   Feature_6  Feature_7  Target
0  -0.452306   0.760415       1
1   0.742095   1.561511       0
2   0.963879  -1.369803       0
3  -1.371901  -0.209324       0
4  -1.065114  -0.186971       0
```

```
Test Sample: [array([-0.90320608,  0.9220511 , -1.32308979,  0.41081065,  1.64201516,
       -1.23559176, -0.63896175,  1.00981709])]
Predicted Class: 1
```

5b. Implement Hidden Markov Models using hmmlearn

Code :

Step 1: Install Required Libraries

 # Install hmmlearn

!pip install hmmlearn

Step 2: Import Required Libraries

 # Import necessary libraries

import numpy as np import pandas as pd from hmmlearn import hmm

import matplotlib.pyplot as plt

Step 3: Create or Load a Dataset

 # Generate synthetic observable data np.random.seed(42)

 # Create a sequence of observations and hidden states

observations = np.random.choice(['A', 'B', 'C'], size=100, p=[0.5, 0.3, 0.2])

hidden_states = np.random.choice(['X', 'Y'], size=100, p=[0.6, 0.4])

 # Save the data in a DataFrame for analysis data = pd.DataFrame({'Observations': observations, 'Hidden States': hidden_states})

print("Generated Data:")

print(data.head())

Step 4: Encode Observations

 # Encode the observations into integers observation_mapping = {obs: idx for idx, obs in enumerate(np.unique(observations))} encoded_observations = np.array([observation_mapping[obs] for obs in observations])

 # Print the mapping print("Observation Encoding:") print(observation_mapping)

Step 5: Initialize and Configure the

HMM

 # Initialize the HMM model n_states = 2 **# Number of hidden states** n_observations = len(observation_mapping) **# Number of unique observations** model = hmm.MultinomialHMM(n_components=n_states, random_state=42, n_iter=100, tol=0.01)

 # Define start probabilities (initial distribution of states)

start_probs = np.array([0.6, 0.4]) # Assumed probabilities

model.startprob_ = start_probs

 # Define transition probabilities between states trans_probs

= np.array([

[0.7, 0.3], # From state X

[0.4, 0.6], # From state Y]) model.transmat_ =

trans_probs

 # Define emission probabilities (probability of observations given states) emission_probs

= np.array([

[0.5, 0.4, 0.1], # State X emits A, B, C

[0.2, 0.3, 0.5], # State Y emits A, B, C

])

model.emissionprob_ = emission_probs

 # Print the configured model parameters print("Start Probabilities:", model.startprob_) print("Transition Matrix:", model.transmat_) print("Emission Probabilities:", model.emissionprob_)

Step 6: Train the Model

 # Reshape the data for HMM (requires 2D array) encoded_observations = encoded_observations.reshape(-1, 1)

 # Fit the model
model.fit(encoded_observations)

 # Predict hidden states for the observations predicted_states = model.predict(encoded_observations) **# Print the predicted states** print("Predicted States:")
print(predicted_states)

Step 7: Visualize the Results

 # Map predicted states back to their original labels

state_mapping = {0: 'X', 1: 'Y'}

predicted_state_labels = [state_mapping[state] for state in predicted_states]

 # Add predicted states to the DataFrame data['Predicted States'] = predicted_state_labels

 # Display the first few rows with predicted states
print("Data with Predicted States:")
print(data.head())
 # Plot the observations and predicted states

plt.figure(figsize=(12, 6))

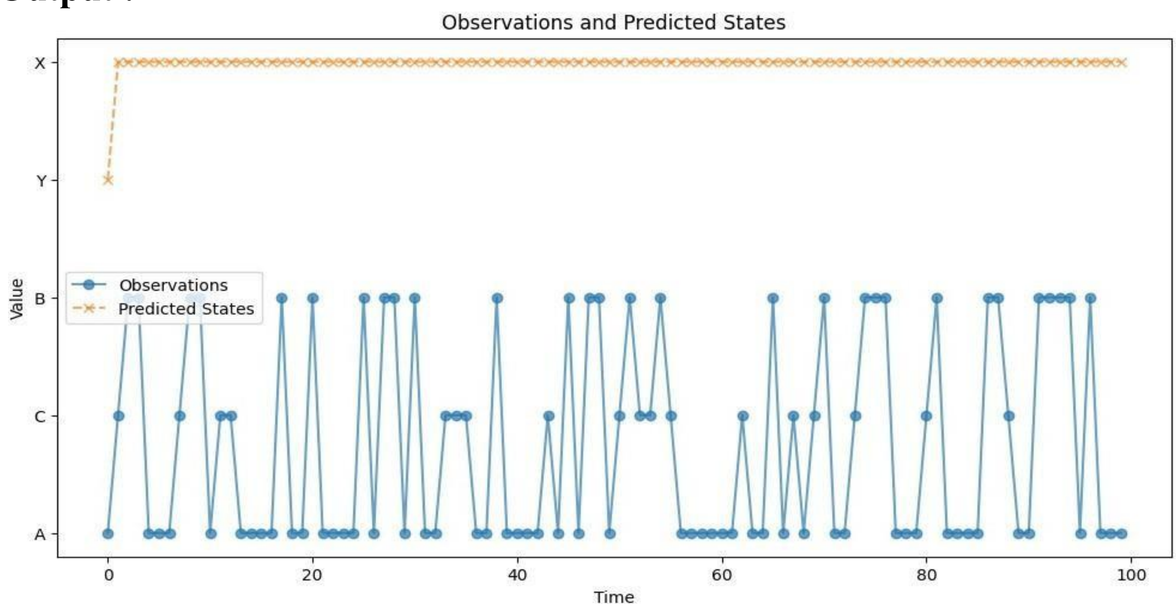
plt.plot(data['Observations'], label='Observations', marker='o', linestyle='-', alpha=0.7)
plt.plot(data['Predicted States'], label='Predicted States', marker='x', linestyle='--', alpha=0.7)
plt.legend()

plt.title("Observations and Predicted

States") plt.xlabel("Time")

plt.ylabel("Value") plt.show()

Output :



Practical 6 : Probabilistic Model

6a. Implement Bayesian Linear Regression to explore prior and posterior distribution.

Bayesian Linear Regression is a probabilistic approach to linear regression that incorporates uncertainty in the model parameters. Instead of estimating point values for parameters (as in traditional linear regression), we estimate distributions over the parameters.

Code :

Step 1: Install Required Libraries

 # Install necessary libraries

!pip install matplotlib seaborn scikit-learn

Step 2: Import Required Libraries

 # Import necessary libraries

import numpy as np import pandas

as pd import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.linear_model import BayesianRidge from sklearn.model_selection import train_test_split from sklearn.metrics import mean_squared_error from google.colab import files

Step 3: Create or Upload a Dataset #

Upload a CSV file if you have one

print("Do you have a CSV file to upload? (yes/no)")
response = input().lower() if response == "yes":

Upload the CSV file

uploaded = files.upload()

filename =

list(uploaded.keys())[0]

else:

 # Generate synthetic data for demonstration
np.random.seed(42)

X = np.random.rand(100, 1) * 10 #

Random data between 0 and 10 y = 2 * X

+ 1 + np.random.randn(100, 1) * 2

 # y = 2x + 1 with some noise #

Convert to a DataFrame

data = pd.DataFrame(np.hstack((X, y)), columns=["X", "y"])
 # Save to CSV for convenience

filename="synthetic_data.csv"

data.to_csv(filename,index=False)

print(f"Synthetic dataset saved as {filename}.")

Step 4: Load and Explore the Data
 # Load the dataset (for CSV file)
data = pd.read_csv(filename) #
Display first few rows
print("Dataset Preview:")
print(data.head())

Step 5: Preprocess the Data

 # Separate features (X) and target (y)

X = data["X"].values.reshape(-1, 1) # Feature matrix y
= data["y"].values # Target vector

 # Split the dataset into training (80%) and testing (20%) sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 6: Implement Bayesian Linear Regression Model

 # Initialize the BayesianRidge model (which implements Bayesian Linear Regression)

bayesian_regressor = BayesianRidge() # **Fit the model on the training data**

bayesian_regressor.fit(X_train, y_train)

 # Predict on the test data y_pred =

bayesian_regressor.predict(X_test)

Step 7: Visualize the Prior and Posterior Distributions # Plot

the prior and posterior distributions of the parameters fig,

ax = plt.subplots(1, 2, figsize=(12, 6))

 # Plot prior distribution (assuming the model starts with a standard prior)

ax[0].set_title("Prior Distribution (Assumed)") ax[0].hist(np.random.normal(0, 1, 1000), bins=50, alpha=0.7, color='blue', label="Prior") ax[0].legend()

 # Plot posterior distribution (after model fitting) ax[1].set_title("Posterior Distribution (After Fitting)")

ax[1].hist(bayesian_regressor.coef_, bins=50, alpha=0.7, color='green', label="Posterior") ax[1].legend()

plt.show()

Step 8: Evaluate the Model Performance # Calculate the Mean Squared Error (MSE)

mse = mean_squared_error(y_test, y_pred) print(f"Mean Squared Error (MSE): {mse:.2f}")

Step 9: Visualize the Fit of the Model

 # Plot the true values and the predicted values

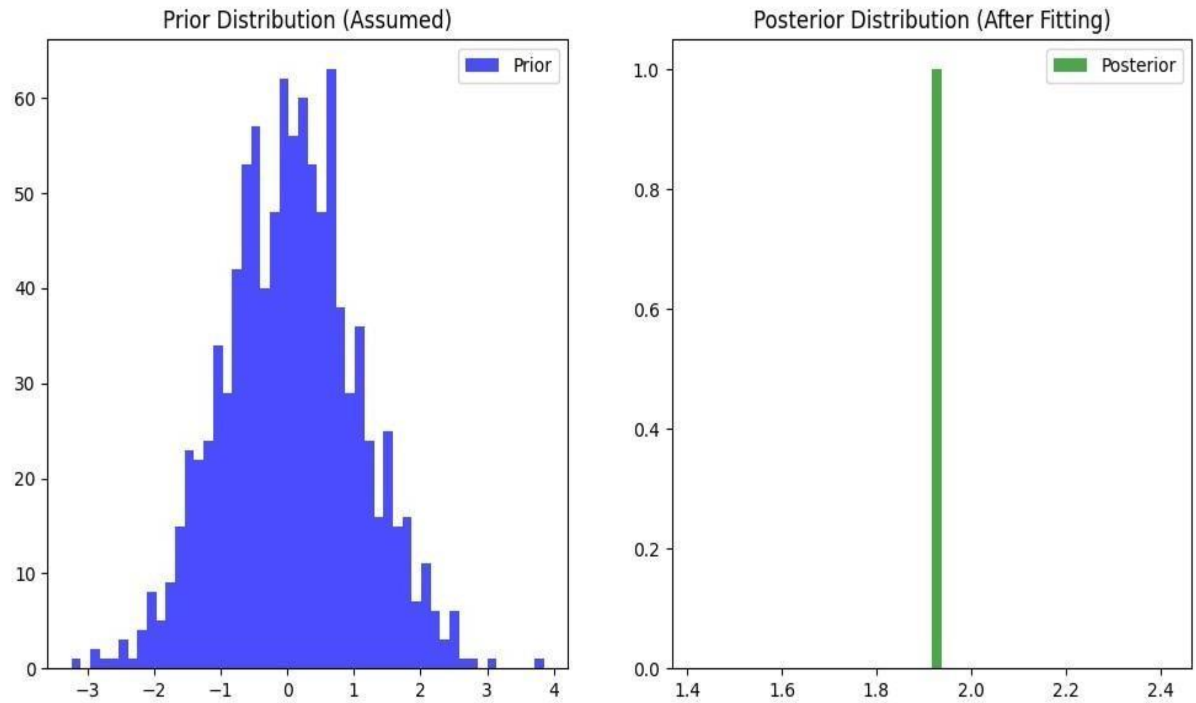
plt.figure(figsize=(8, 6))

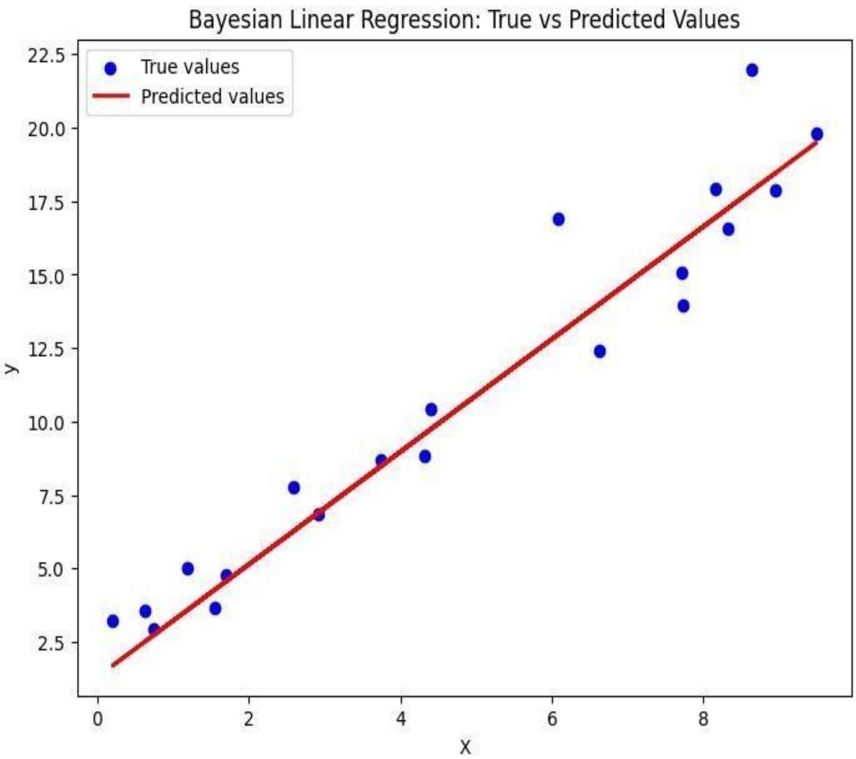
plt.scatter(X_test, y_test, color="blue", label="True values")

plt.plot(X_test, y_pred, color="red", label="Predicted values", linewidth=2) plt.title("Bayesian Linear Regression: True vs Predicted Values") plt.xlabel("X") plt.ylabel("y") plt.legend() plt.show()

Mean Squared Error (MSE): 3.9

Output :





6b. Implement Gaussian Mixture Models for density estimation and unsupervised clustering.

Code :

Step 1: Install Required Libraries

 # Install required libraries

!pip install matplotlib seaborn scikit-learn

Step 2: Import Required Libraries

 # Import necessary libraries

import numpy as np import pandas

as pd import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.mixture import GaussianMixture from sklearn.model_selection import train_test_split from google.colab import files

Step 3: Create or Upload a Dataset

#Ask if the user has a CSV file to upload print("Do

you have a CSV file to upload? (yes/no)") response

= input().lower() if response == "yes": **# Upload the**

CSV file uploaded = files.upload() filename =

list(uploaded.keys())[0] else:

 # Generate synthetic 2D data with two clusters for demonstration np.random.seed(42)
 # Generate data for two Gaussian distributions

X1 = np.random.normal(loc=0, scale=1, size=(300, 2)) # Cluster 1: mean = 0, std = 1

X2 = np.random.normal(loc=5, scale=1, size=(300, 2)) # Cluster 2: mean = 5, std = 1 **# Stack the data to create a dataset**

X = np.vstack([X1, X2])
 # Create DataFrame to simulate the CSV file for consistency
data = pd.DataFrame(X, columns=["Feature_1", "Feature_2"])
filename = "synthetic_data.csv" data.to_csv(filename, index=False) print(f"Synthetic dataset saved as {filename}.")

Step 4: Load and Explore the Dataset

 # Load the dataset (if CSV file is uploaded)
data = pd.read_csv(filename) **# Display the first few rows** print("Dataset Preview:")
print(data.head())

 # Plot the data to visualize its structure
sns.scatterplot(data=data, x="Feature_1",

y="Feature_2") plt.title("Synthetic Data")

plt.xlabel("Feature 1") plt.ylabel("Feature 2") plt.show()

Step 5: Fit a Gaussian Mixture Model (GMM)

 # Define the GMM model

n_components = 2 **# Number of Gaussian distributions (clusters)** gmm = GaussianMixture(n_components=n_components, covariance_type='full', random_state=42)

 # Fit the GMM model to the data gmm.fit(data)

 # Predict the cluster labels for each data point labels

= gmm.predict(data)

 # Add the cluster labels to the dataset for visualization
data['Cluster'] = labels **# Plot the clustered data**

sns.scatterplot(data=data, x="Feature_1", y="Feature_2", hue="Cluster", palette="viridis", marker="o")

plt.title("Gaussian Mixture Model Clustering")

plt.xlabel("Feature 1") plt.ylabel("Feature 2")

plt.legend()

plt.show()

Step 6: Visualize the Gaussian Mixture Model (GMM) Components
 # Extract the means and covariances of the Gaussian components
means = gmm.means_ covariances = gmm.covariances_

 # Plot the GMM components on top of the data plt.figure(figsize=(8, 6)) **# Plot data points** sns.scatterplot(data=data, x="Feature_1", y="Feature_2", hue="Cluster", palette="viridis", marker="o", s=60, alpha=0.7)

 # Plot the GMM ellipses for mean, covar in zip(means, covariances):

 # Plot the Gaussian components as ellipses

v, w = np.linalg.eigh(covar) v = 2.0 *

np.sqrt(2.0) * np.sqrt(v)

Scaling factor for the ellipse

u = w[0] / np.linalg.norm(w[0])

 # Normalize the eigenvector

angle = np.arctan(u[1] / u[0])

 # Create the ellipse angle = angle * 180.0 / np.pi #

Convert to degrees

ellipse = plt.matplotlib.patches.Ellipse(mean, v[0], v[1], angle=angle, color='red', alpha=0.3)
plt.gca().add_patch(ellipse)

plt.title("GMM Clustering with Gaussian Components")

plt.xlabel("Feature 1") plt.ylabel("Feature 2")

plt.legend()

plt.show()

Step 7: Model Evaluation (Optional)

 # Compute the log-likelihood of the data under the fitted GMM model log_likelihood
= gmm.score(data)
print(f"Log-Likelihood of the data: {log_likelihood:.2f}")
Step 8: Predict New Data Points

 # Example of predicting the cluster for new data points

new_data = np.array([[1.5, 2.5], [4.5, 5.5], [7.0, 8.0]])

new_labels = gmm.predict(new_data)

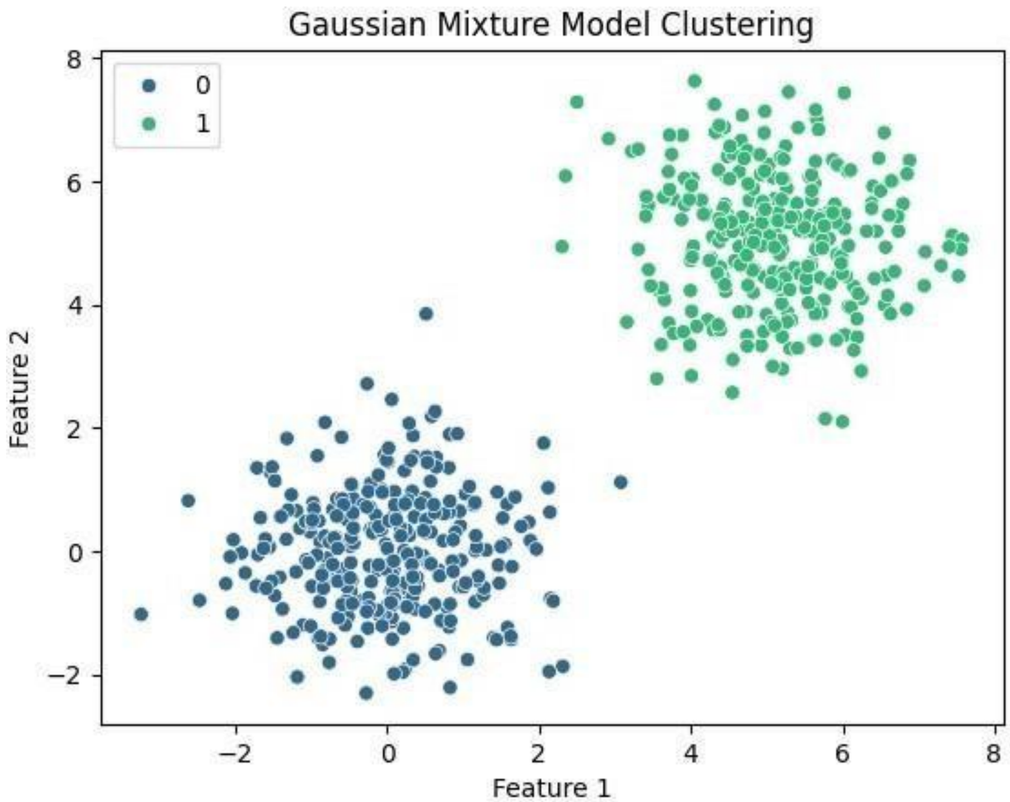
 # Print the predicted clusters for the new data

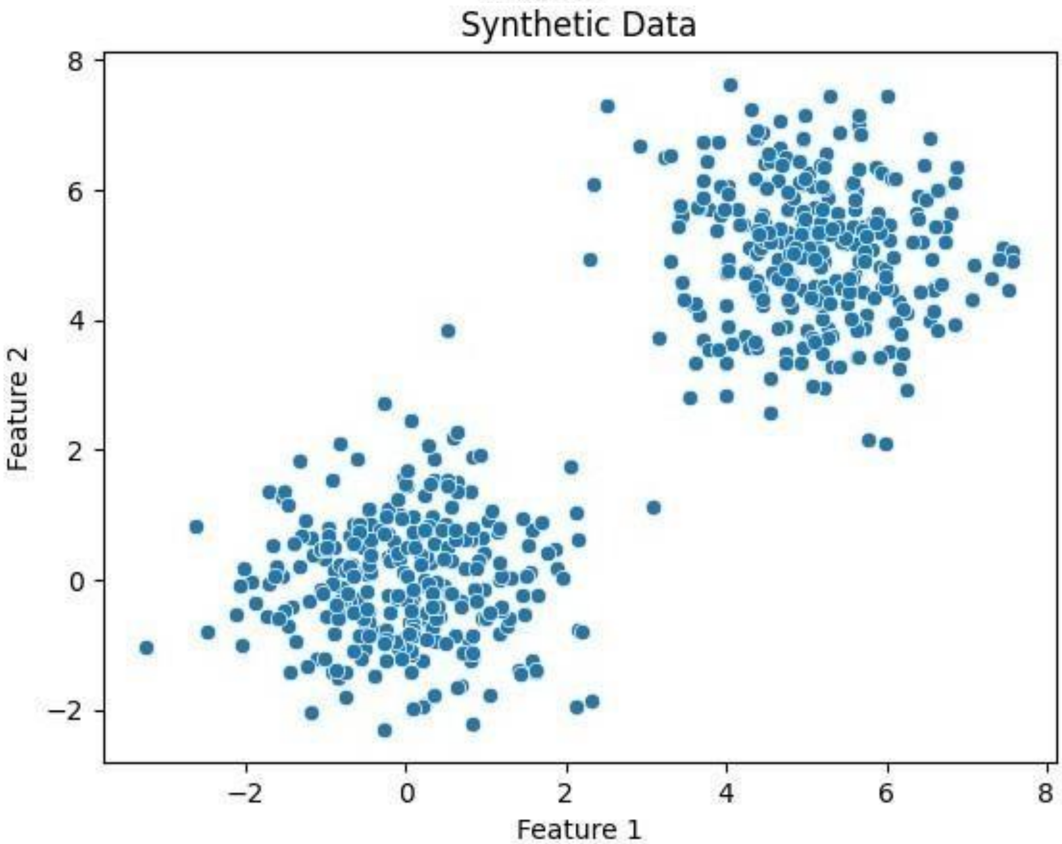
points print("Predicted Clusters for New Data

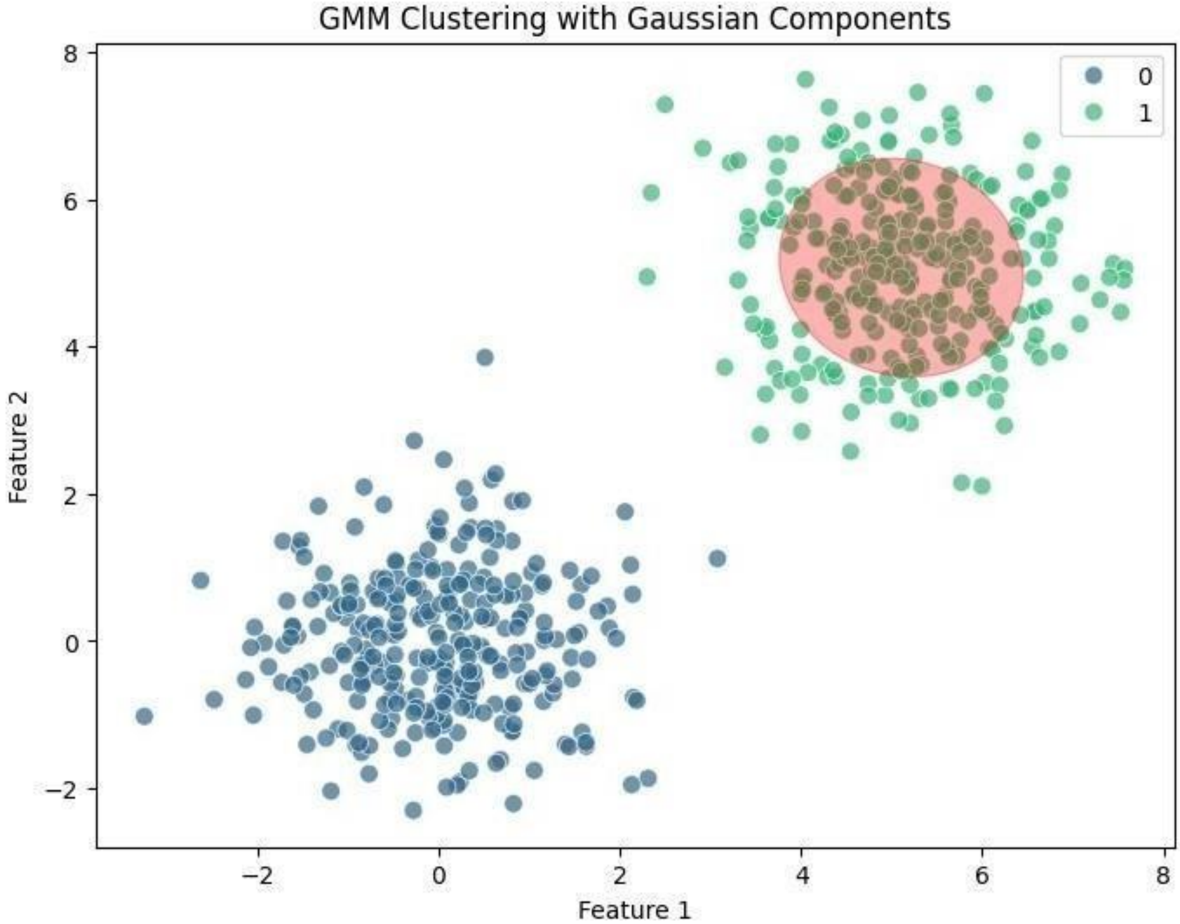
Points:") for i, label in enumerate(new_labels):

print(f"Data point {new_data[i]} is in Cluster {label}")

Output :


Gaussian Mixture Model Clustering
0
1
Feature 2
Feature 1
8
6
4
2
0
−2
−2
0
2
4
6
8


Synthetic Data
Feature 2
Feature 1
8
6
4
2
0
−2
−2
0
2
4
6
8



Practical 7 : Model Evaluation and Hyperparameter Tuning

7a. Implement cross-validation techniques (k-fold, stratified, etc.) for robust model evaluation

Code :

1. Import Necessary Libraries import numpy as np import pandas as pd

from sklearn.datasets import make_classification from sklearn.model_selection import train_test_split, KFold, StratifiedKFold, GridSearchCV from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import matplotlib.pyplot as plt import seaborn as sns

2. Generate a Synthetic Dataset

 # Create a synthetic dataset with 2 classes

X, y = make_classification(n_samples=1000, n_features=10,

n_informative=8, n_redundant=2, n_clusters_per_class=1,

random_state=42

)

 # Convert to a DataFrame for visualization

df = pd.DataFrame(X, columns=[f'Feature_{i}' for i in range(1, 11)]) df['Target']

= y

 # Display the first few rows

print(df.head())

3. Split Data into Train and Test Sets

 # Split data into 80% training and 20% testing

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

4. Define k-Fold Cross-Validation kf = KFold(n_splits=5, shuffle=True, random_state=42) print("k-Fold Cross-Validation:") for train_index, val_index in kf.split(X_train): print("TRAIN:", train_index, "VALIDATION:", val_index)

5. Define Stratified k-Fold Cross-Validation skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42) print("\nStratified k-Fold Cross-Validation:") for train_index, val_index in skf.split(X_train, y_train): print("TRAIN:", train_index, "VALIDATION:", val_index)

6. Train and Evaluate Using k-Fold Cross-Validation # Initialize model

model = RandomForestClassifier(random_state=42)

```
# Perform k-Fold Cross-Validation
accuracies = [] for train_index, val_index in
kf.split(X_train):
    X_kf_train, X_kf_val = X_train[train_index], X_train[val_index] y_kf_train,
    y_kf_val = y_train[train_index], y_train[val_index]
    # Train model
    model.fit(X_kf_train, y_kf_train)
    # Validate model
    y_pred = model.predict(X_kf_val) accuracy =
accuracy_score(y_kf_val, y_pred) accuracies.append(accuracy)
print(f"Average Accuracy from k-Fold: {np.mean(accuracies):.2f}")
```

7. Hyperparameter Tuning Using GridSearchCV

```
# Define parameter grid param_grid = {
    'n_estimators': [50, 100, 200],
    'max_depth': [None, 10, 20, 30],
    'min_samples_split': [2, 5, 10],
}
# Perform GridSearchCV with Stratified k-Fold grid_search
= GridSearchCV(
    estimator=RandomForestClassifier(random_state=42),
    param_grid=param_grid,
    cv=StratifiedKFold(n_splits=5, shuffle=True,
    random_state=42), scoring='accuracy', n_jobs=-1, verbose=1
)
# Fit to training data grid_search.fit(X_train,
y_train)
print("Best Parameters:", grid_search.best_params_)
print("Best Cross-Validation Accuracy:", grid_search.best_score_)
```

8. Evaluate the Final Model # Use the

best model for evaluation best_model =

grid_search.best_estimator_

 # Predict on test data

y_test_pred = best_model.predict(X_test)

 # Evaluate performance print("\nTest Accuracy:", accuracy_score(y_test,

y_test_pred)) print("\nClassification Report:\n", classification_report(y_test,

y_test_pred)) **# Confusion matrix**

conf_matrix = confusion_matrix(y_test, y_test_pred)

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues', xticklabels=['Class 0', 'Class 1'], yticklabels=['Class 0', 'Class 1'])

plt.xlabel('Predicted')

plt.ylabel('Actual')

plt.title('Confusion Matrix')

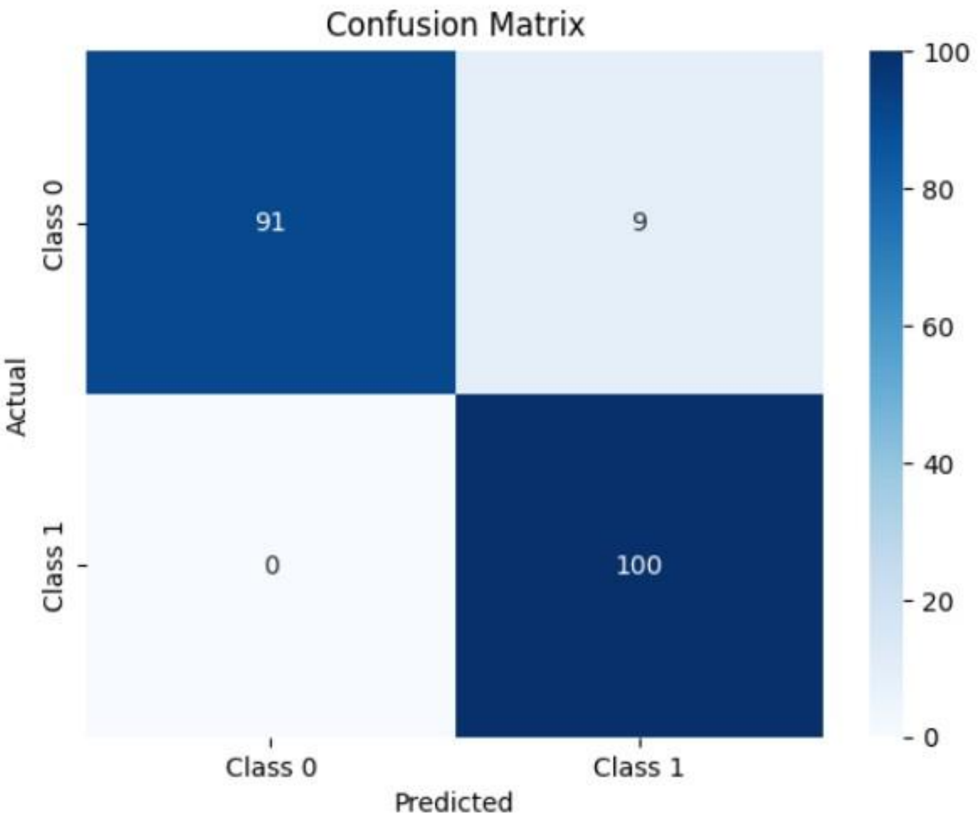
plt.show()

Output :

```
   Feature_1  Feature_2  Feature_3  Feature_4  Feature_5  Feature_6  \
0  -3.358483   3.159918   0.827163   0.069638  -6.715639  -2.708559
1   2.071819  -4.055419  -2.615940  -2.599432   3.053752   0.366795
2  -0.633460   0.712482   2.024390  -0.432639  -1.307929   0.419320
3  -0.464478   0.892442   2.521010   2.766580  -1.933734  -1.418018
4   1.042426  -1.192605  -2.071386  -0.131231   0.545377   0.379060

   Feature_7  Feature_8  Feature_9  Feature_10  Target
0   0.183206   1.113502   1.730759    1.228394       1
1  -0.392171  -1.191720  -1.220516    1.899925       0
2  -1.469510  -0.719051   1.155005    2.018026       0
3   1.391760  -2.430279   1.308295   -0.270896       1
4  -0.062978  -1.325591   2.037936    0.115414       0
k-Fold Cross-Validation:
TRAIN: [  0   1   3   4   5   6   8   9  11  12  13  14  15  16  17  18  19  20
  21  22  24  25  26  27  28  32  34  35  36  37  38  40  41  42  43  44
  45  46  47  48  50  51  52  53  55  56  57  58  59  60  61  62  64  68
  69  70  71  73  74  75  79  80  82  83  85  87  88  89  90  91  92  93
  94  95  98  99 100 102 103 104 105 106 107 108 111 112 113 114 115 116
 117 119 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136
 138 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 156 157
 158 159 160 161 162 163 164 165 166 167 169 170 171 172 173 175 176 177
 178 179 180 181 182 183 184 185 186 187 188 189 190 191 193 194 195 196
 197 200 201 202 203 205 206 207 212 213 214 216 217 219 220 221 222 223
 224 225 226 227 228 229 230 232 233 234 236 237 238 239 240 241 242 243
 245 246 247 248 249 251 252 253 255 256 257 258 259 261 262 263 264 267
 268 269 270 271 272 273 274 276 277 278 279 280 282 283 284 285 287 288
 289 290 291 292 293 295 297 298 299 300 301 303 304 305 307 308 309 310
 311 312 313 315 317 318 319 320 321 322 324 325 328 329 330 331 332 334
```

```
 674 675 676 677 678 679 680 681 682 683 684 685 686 687 689 690 691 693
 694 695 697 698 699 700 701 702 703 704 706 707 708 709 710 711 712 713
 714 716 717 718 722 724 725 726 727 728 729 730 731 732 733 734 735 736
 737 739 740 742 743 745 747 748 749 751 752 753 755 756 757 759 761 762
 763 765 766 767 768 769 770 771 772 773 774 775 778 779 780 781 782 784
 785 788 789 790 791 792 793 794 797 799] VALIDATION: [  2   7  10  23  29  30  31  33  39  49  54  63  65  66  67  72  76  77
  78  81  84  86  96  97 101 109 110 118 120 137 139 155 168 174 192 198
 199 204 208 209 210 211 215 218 231 235 244 250 254 260 265 266 275 281
 286 294 296 302 306 314 316 323 326 327 333 336 346 352 357 360 361 365
 367 368 377 383 388 393 395 398 409 422 423 425 428 432 446 456 464 481
 483 486 490 506 512 513 515 519 521 525 526 529 532 533 534 537 545 568
 570 578 589 591 594 595 596 604 608 610 621 622 628 635 637 640 641 644
 652 655 656 658 662 666 667 670 688 692 696 705 715 719 720 721 723 738
 741 744 746 750 754 758 760 764 776 777 783 786 787 795 796 798]
TRAIN: [  0   1   2   3   4   5   7   8   9  10  12  13  14  15  16  17  19  20
  21  22  23  25  26  27  29  30  31  32  33  34  35  36  37  38  39  40
  45  46  47  48  49  50  52  53  54  57  58  59  62  63  64  65  66  67
  68  71  72  75  76  77  78  80  81  84  85  86  87  88  91  93  94  95
  96  97  98  99 100 101 102 103 105 106 107 109 110 111 112 113 115 116
 117 118 119 120 121 122 123 124 125 126 127 128 129 130 134 137 138 139
 141 142 143 144 146 147 149 150 151 152 153 154 155 156 157 159 160 161
 162 166 168 169 170 171 172 173 174 175 179 180 183 184 185 186 187 188
 189 190 191 192 194 195 197 198 199 200 201 202 203 204 205 206 207 208
 209 210 211 214 215 216 217 218 219 221 222 224 225 226 228 229 230 231
 232 233 235 236 237 238 240 241 242 243 244 245 246 249 250 251 252 253
 254 255 256 257 258 260 261 262 263 265 266 267 268 269 270 271 272 273
 274 275 276 277 278 279 280 281 282 283 284 286 287 288 289 293 294 295
 296 297 298 301 302 303 304 305 306 307 308 310 312 313 314 315 316 317
 318 320 321 322 323 324 325 326 327 330 333 335 336 337 338 339 340 341
```

```
 450 455 459 460 463 470 472 475 480 481 483 494 496 502 503 505 514 519
 521 523 529 533 553 555 560 563 565 579 585 590 594 600 603 620 630 631
 634 635 637 638 644 646 648 649 651 673 675 686 691 693 708 709 715 720
 729 730 738 747 753 759 767 771 774 776 777 780 782 783 784 796]
Average Accuracy from k-Fold: 0.95
Fitting 5 folds for each of 36 candidates, totalling 180 fits
Best Parameters: {'max_depth': None, 'min_samples_split': 5, 'n_estimators': 50}
Best Cross-Validation Accuracy: 0.96

Test Accuracy: 0.955

Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.91      0.95       100
           1       0.92      1.00      0.96       100

    accuracy                           0.95       200
   macro avg       0.96      0.96      0.95       200
weighted avg       0.96      0.95      0.95       200
```


Confusion Matrix
Actual
Class 0
Class 1
91
9
0
100
Class 0
Class 1
Predicted
100
80
60
40
20
0

7b. Systematically explore combinations of hyperparameters to optimize model performance.(use grid and randomized search)

Code :

1. Import Necessary Libraries import numpy as np import pandas

as pd

from sklearn.datasets import make_classification

from sklearn.model_selection import train_test_split, GridSearchCV, RandomizedSearchCV, StratifiedKFold

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix import

matplotlib.pyplot as plt

import seaborn as sns

2. Generate a Synthetic Dataset

 # Generate a binary classification dataset

X, y = make_classification(n_samples=1000, n_features=12,

n_informative=8, n_redundant=2, n_clusters_per_class=1,

flip_y=0.03, random_state=42

)

 # Convert to a DataFrame for visualization

df = pd.DataFrame(X, columns=[f'Feature_{i}' for i in range(1, 13)]) df['Target']

= y

 # Display the first few rows

print(df.head())

3. Split Data into Train and Test Sets

 # Split data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

4. Define the Model

 # Initialize a Random Forest classifier

model = RandomForestClassifier(random_state=42)

5. Hyperparameter Tuning Using Grid Search # Define a parameter grid for Grid Search param_grid = {

'n_estimators': [50, 100, 200],

'max_depth': [None, 10, 20],

'min_samples_split': [2, 5, 10],

'min_samples_leaf': [1, 2, 4]

}

 # GridSearchCV with 5-fold cross-validation

grid_search = GridSearchCV(

estimator=model, param_grid=param_grid,

scoring='accuracy',

cv=StratifiedKFold(n_splits=5, shuffle=True,

random_state=42), verbose=1, n_jobs=-1

)

 # Fit the model

grid_search.fit(X_train, y_train)

 # Best parameters and score from Grid Search

print("Best Parameters from Grid Search:", grid_search.best_params_)

print("Best Cross-Validation Accuracy from Grid Search:", grid_search.best_score_)

6. Hyperparameter Tuning Using Randomized Search

from scipy.stats import randint

 # Define a parameter distribution for Randomized Search

param_dist = {

'n_estimators': randint(50, 300),

'max_depth': [None, 10, 20, 30],

'min_samples_split': randint(2, 15),

'min_samples_leaf': randint(1, 10)

}

 # RandomizedSearchCV with 5-fold cross-validation

random_search = RandomizedSearchCV(

estimator=model,

param_distributions=param_dist,

n_iter=50, # Number of random combinations to try scoring='accuracy',

cv=StratifiedKFold(n_splits=5, shuffle=True,

random_state=42), verbose=1, n_jobs=-1, random_state=42

)

 # Fit the model

random_search.fit(X_train, y_train)

 # Best parameters and score from Randomized Search

print("Best Parameters from Randomized Search:", random_search.best_params_)

print("Best Cross-Validation Accuracy from Randomized Search:", random_search.best_score_)

7. Evaluate the Best Model

Select the best model from Grid Search and Randomized Search

best_model = random_search.best_estimator_ # Or use grid_search.best_estimator_

 # Predict on test data

y_test_pred = best_model.predict(X_test)

 # Evaluate the performance

print("\nTest Accuracy:", accuracy_score(y_test, y_test_pred))

print("\nClassification Report:\n", classification_report(y_test, y_test_pred))

 # Confusion Matrix

conf_matrix = confusion_matrix(y_test, y_test_pred)

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues', xticklabels=['Class 0', 'Class 1'], yticklabels=['Class 0', 'Class 1'])

plt.xlabel('Predicted')

plt.ylabel('Actual')

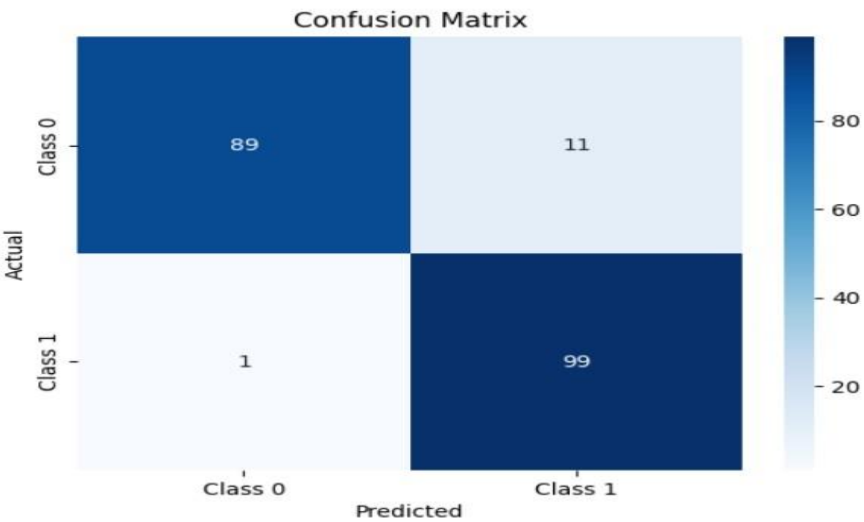
plt.title('Confusion Matrix')

plt.show()

Output :

```
   Feature_1  Feature_2  Feature_3  Feature_4  Feature_5  Feature_6  \
0   0.013650   0.607473  -2.096916   2.867232   2.504360   0.784101
1   0.107199   0.105735  -3.843343   1.524052  -1.619824   0.778334
2  -1.779086  -5.219831  -0.738488   2.108084  -0.803833  -3.431122
3  -4.310656  -2.268569   1.864943  -1.246116   1.260794  -2.007664
4  -3.195179  -0.671327   3.720485   0.356661   0.819486   2.670230

   Feature_7  Feature_8  Feature_9  Feature_10  Feature_11  Feature_12  Target
0  -0.497744  -0.482072   1.112773    1.641637   -2.689832   -0.480311       1
1   0.551177  -1.843583  -0.110132   -0.494739   -0.985276   -0.978400       0
2   1.346120  -0.858351  -0.792415   -2.260815    0.238780    3.029952       0
3  -0.824133  -2.277449   0.936206    1.255903    1.386278   -0.321200       0
4   1.857477  -3.410944  -1.773719    0.656476    3.534189   -1.704889       0
Fitting 5 folds for each of 81 candidates, totalling 405 fits
Best Parameters from Grid Search: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}
Best Cross-Validation Accuracy from Grid Search: 0.9487499999999999
Fitting 5 folds for each of 50 candidates, totalling 250 fits
Best Parameters from Randomized Search: {'max_depth': None, 'min_samples_leaf': 2, 'min_samples_split': 9, 'n_estimators': 285}
Best Cross-Validation Accuracy from Randomized Search: 0.9487499999999999

Test Accuracy: 0.94
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.89      0.94       100
           1       0.90      0.99      0.94       100

    accuracy                           0.94       200
   macro avg       0.94      0.94      0.94       200
weighted avg       0.94      0.94      0.94       200
```



Practical 8 : Bayesian Learning

Implement Bayesian Learning using inferences

Code :

1. **Import Necessary Libraries** import numpy as np import pandas as pd

from sklearn.datasets import make_classification

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix import seaborn as sns

import matplotlib.pyplot as plt

2. Generate a Synthetic Dataset

We create a dataset suitable for classification problems.

 # Generate a dataset with 2 classes

X, y = make_classification(n_samples=1000, n_features=8, n_informative=6, n_redundant=2, n_classes=2, random_state=42) # **Convert to DataFrame for visualization** df = pd.DataFrame(X, columns=[f'Feature_{i}' for i in range(1, 9)]) df['Target'] = y

 # Display the first few rows

print(df.head())

3. Split the Dataset

Divide the data into training and testing sets.

 # Split data into 80% training and 20% testing

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

4. Bayesian Learning with Naive Bayes

Here, we implement Bayesian Learning using the Gaussian Naive Bayes classifier.

 # Initialize the Gaussian Naive Bayes model model = GaussianNB()

 # Fit the model to the training data

model.fit(X_train, y_train) # **Predict on the test data**

y_pred = model.predict(X_test)

5. Evaluate the Model

We evaluate the model's performance using accuracy, classification report, and confusion matrix.

Calculate accuracy accuracy =

accuracy_score(y_test, y_pred)

print(f"Test Accuracy: {accuracy:.2f}")

 # Print classification report

print("\nClassification Report:\n", classification_report(y_test, y_pred))

 # Generate and plot confusion matrix conf_matrix

= confusion_matrix(y_test, y_pred)

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues', xticklabels=['Class 0', 'Class 1'], yticklabels=['Class 0', 'Class 1'])

plt.xlabel('Predicted')

plt.ylabel('Actual')

plt.title('Confusion Matrix')

plt.show()

6. Understanding Bayesian Inference

In Bayesian Learning, the model predicts based on the probabilities:

- **Prior Probability (P(C)P(C)P(C)):** The likelihood of each class based on historical data.
- **Likelihood (P(X | C)P(X|C)P(X | C)):** The probability of the data given a class.
- **Posterior Probability (P(C | X)P(C|X)P(C | X)):** Calculated using Bayes' theorem: P(C|X)=P(X|C)·P(C)P(X)P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}P(C|X)=P(X)P(X|C)·P(C)

 # Example: Compute posterior probabilities for the first test sample

sample = X_test[0].reshape(1, -1) posterior_probs =

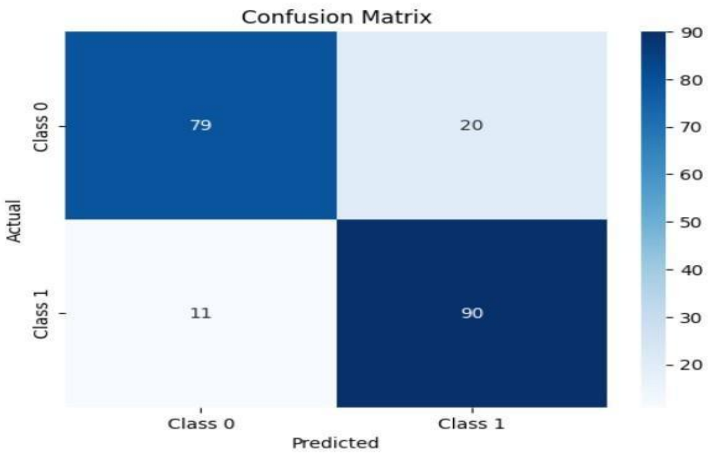
model.predict_proba(sample) print(f"Sample Features: {sample}")

print(f"Posterior Probabilities: {posterior_probs}")

print(f"Predicted Class: {model.predict(sample)}")

Output :

```
   Feature_1  Feature_2  Feature_3  Feature_4  Feature_5  Feature_6  \
0  -1.732538   5.260112  -2.952194  -4.603768   2.235848   1.928893
1   2.072914   2.240572  -1.385104  -2.514962  -0.984756   1.436260
2  -0.263106   1.527781  -1.872414  -0.028009   1.612809   3.264194
3  -0.164349  -0.550131  -0.019503  -0.765000   2.273523   2.084217
4  -1.419423   1.015324  -0.864441  -0.009297   0.385404   0.449093

   Feature_7  Feature_8  Target
0  -0.101845   3.193487       0
1  -1.255271   2.089872       0
2  -1.296421   1.537870       0
3  -0.321931   0.426253       0
4  -0.029007  -1.902917       1
Test Accuracy: 0.84

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.80      0.84        99
           1       0.82      0.89      0.85       101

    accuracy                           0.84       200
   macro avg       0.85      0.84      0.84       200
weighted avg       0.85      0.84      0.84       200
```




```
Sample Features: [[ 1.94397502  0.33753289 -0.77968371  2.63667244 -1.62323864  1.87859635
  -2.20522563  2.09632056]]
Posterior Probabilities: [[0.07067404 0.92932596]]
Predicted Class: [1]
```